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# Derivative Lasso

## Credibility-based signal fitting for GLMs

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## Abstract

Building transparent and accurate models for insurance risks is challenging due to the complex correlations and nonlinearities inherent in the modeled effects. Generalized Linear Models (GLMs) are an essential tool for addressing these correlations and building transparent, table-based models. Nevertheless, capturing nonlinear effects in GLMs requires a lot of iterative work — so does developing robust and credible models.

This whitepaper posits that these limitations can be effectively addressed through a penalized regression framework, while preserving the model parametrization and table-based output structure of GLMs. The approach is statistically sound, as it rests on assumptions that intuitively integrate credibility theory and nonlinear effects into the modeling process. Examples drawn from publicly available data are provided throughout this paper in order to guide insurance practitioners through the construction and analysis of such models.

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# Introduction

Actuaries apply mathematical and statistical methods to provide sound estimates of risk. For the past two decades, the de-facto standard statistical methodology has been Generalized Linear Models (GLMs). GLMs have gained widespread popularity due to their ability to explicitly define statistical assumptions in data, model correlations among variables, and directly generate output in the form of a rating table.

Nonetheless, GLMs are not without their limitations. They exhibit instability when applied to segments with limited data (often referred to as a "lack of credibility") and involve multiple layers of complexity in capturing nonlinear relationships between variables and risk. The incapacity to inherently account for nonlinear effects is a major constraint, given that nonlinearities are commonly observed in insurance data. While techniques like feature engineering and variable transformation can partially alleviate this limitation, they require a substantial amount of time-consuming, iterative work and expert judgment, which can be challenging to formalize.

In recent years, there has been a surge in the adoption of other predictive modeling techniques from the field of machine learning, such as Gradient Boosting Machines (GBMs) and Random Forests. These techniques inherently possess the ability to model nonlinearities and manage low-credibility segments within the data, often achieving this in a fraction of the time required to develop an equivalent GLM. Unfortunately, the models produced by these techniques lack transparency and interpretability: they cannot be directly visualized or expressed as rating tables, and modelers can only gain a partial understanding of their behavior through post-hoc analysis. The absence of transparency in these models can make it challenging, if not impossible, for actuaries to interact with and adjust the models to incorporate selections and actuarial judgment.

This leaves the actuary in an unsatisfactory modeling situation. A modeler can create transparent GLMs, but this comes at a significant cost in terms of time and expertise, and often, it may compromise the model's predictiveness. Moreover, there is a relatively high risk of operational errors due to the numerous manual operations involved. Alternatively, a modeler might opt to automate the model-building process using non-transparent techniques. However, this approach can limit the deployability of models in production and results in only a partial understanding of the relationship between explanatory variables and predictions.

This paper proposes a solution to the aforementioned dilemma faced by modelers using the **penalized regression** framework. We will illustrate how **lasso** allows GLMs to include credibility considerations, and how the **derivative lasso**, a variation of the lasso penalty, natively models nonlinearities. The modeling technique preserves the standard GLM parameterization and results in only minor adjustments to the GLM optimization formula.

The proposed framework is designed to address real-world applications with insurance data. To illustrate this, the paper focuses on an actuarial application based on real, publicly available

insurance data. By directly observing how the limitations of GLMs affect the quality of a model, we will employ mathematical concepts to address these limitations and assess how adjustments can enhance both the efficiency of GLM construction and their ability to generalize.

The paper is organized as follows:

- Chapter 1 showcases standard techniques commonly used by practitioners to model nonlinearities within the GLM framework. Using real data examples, we will emphasize the two primary limitations of GLMs: the lack of credibility and inability to natively detect nonlinear effects.
- Chapter 2 illustrates how penalized regression can effectively incorporate credibility into GLMs. We will highlight the impact of the techniques of lasso and ridge on GLM estimates and provide theoretical insights into how penalized regression serves as an efficient means to generalize standard credibility procedures to multivariate modeling.
- Chapter 3 focuses on how the derivative lasso penalty can natively integrate nonlinear effects into GLMs without the need for extensive manual transformations. We begin by offering a theoretical rationale to justify the choice of the penalty structure. Then, we provide an extensive comparison with GLM techniques to illustrate how this penalty significantly enhances existing GLM procedures presented in Chapter 1, such as binning or polynomial transformations.
- Chapter 4 offers practical guidance and best practices for defining, constructing, and evaluating models built using the proposed penalized regression framework. These procedures and best practices have been refined over time through extensive applications of the proposed methodology across a broad spectrum of use-cases in non-life insurance.
- Chapter 5 provides a comparative analysis of the derivative lasso technique against another popular nonlinear modeling technique, i.e., the technique of Generalized Additive Models (spline GAMs).

# 1. Modeling nonlinearities in GLMs

Nonlinearities are a prevalent phenomenon within Property and Casualty (P&C) data. One example is given by the univariate relationship between driver age and accident frequency, for which a nonlinear effect is very clear for young drivers. Typically, the frequency of accidents decreases sharply with time for young drivers, before stagnating between 25 and 40 years old, then the data often shows an increase in frequency with a maximum around 50 years old, when insured drivers' childrens start borrowing their cars. Finally, the frequency of accidents often increases again for older drivers. While these effects vary from one population to another, the strength and strongly nonlinear nature of the relation between driver age and claims frequency is universal in motor insurance.

Despite this, the insurance industry standard for modeling these nonlinear relationships is through the use of Generalized *Linear* Models (GLMs). At first glance, this may seem counterintuitive, as GLMs seem by definition appropriate to model only linear relationships between variables. However, a toolbox of techniques allows GLMs to incorporate nonlinear relationships. The underlying principle is to "make the data-relations linear" through the use of various transformations applied to explanatory variables. Several books provide a comprehensive overview of the theory and the practical strategies to capture nonlinear relationships using GLMs, e.g., see Goldburd et al. (2016) and Wüthrich and Merz (2023).

This chapter aims to highlight the main practical strategies used in GLM modeling, by walking through a practical example:

- **Linear effects modeling** - to capture effects of numeric variables that are linearly related to the response variable.
- **Categorical effects modeling** - to capture effects of non-numeric, non-ordered variables with GLMs.
- **Grouping of levels** - to capture effects of numeric variables that are not linearly related to the response variable.
- **Polynomial effects modeling** - to capture effects of numeric variables that are not linearly related to the response variable.

By analyzing the claim history of an insurance portfolio, we will develop the intuition of why the GLM framework suffers from two main shortcomings [Goldburd et al. (2016)]:

- **Credibility:** Full credibility is given to the data for each coefficient, with no regard to the thinness of the exposure on which the estimate is based.
- **Nonlinearity:** Predictions must be based on a linear function of the predictors.

By understanding the root causes and how these shortcomings impact the GLM modeling procedure, it will be possible to recognize how a proper use of penalized regression

techniques can respectively address credibility concerns (Chapter 2) and incorporate nonlinear effects (Chapter 3).

## 1.1 Claims modeling: A practical example

Throughout the following sections, we will use the **freMtpl** data from Casdatasets package of R Charpentier (2014). This French personal auto database consists of 678,103 records and contains 9 policy attributes. A comprehensive analysis of the database is provided in Noll, Salzmann, and Wuthrich (2020) and Wüthrich and Merz (2023). The objective of the modeling exercise is the prediction of frequency of claims based on policy attributes given observed historical observations. More specifically, we define the frequency as the expected number of claims (called **ClaimNb** in the database) per exposure. This exposure value is expressed in years in the **Exposure** column of this database.

The model should aim to select the combination of policy attributes that best explain the observed risk. In the case of two possible models with almost equivalent results, the modeler should always favor the model that uses the least amount of policy information (according to the **parsimony principle**).

Practitioners with experience in GLM modeling are well aware that using the raw claim data as an input of a GLM solver will rarely build an acceptable model. An appropriate GLM is often the result of a large number of iterations, discoveries and tests. We will highlight practically **why** all these iterations, despite being tedious and error-prone, are required for the practitioner. Once these reasons are well defined, then it will be possible to suggest solutions that can effectively address these shortcomings.

To develop an appropriate model, we will proceed by:

- Modeling some of the policy attributes individually (i.e., **univariate analysis**) to predict the claim frequency;
- Analyzing how the resulting relationships change when all policy factors are included simultaneously (**multivariate analysis**).

## 1.2 Univariate analysis

### The Area variable and linear associations

The first variable we will consider in the analysis is the Area variable, which describes the density of the city community where the car driver lives in: from 0 for rural areas to 5 for urban

centers[3]. Figure 1.1 shows the relationship between the Area variable and the claim frequency is

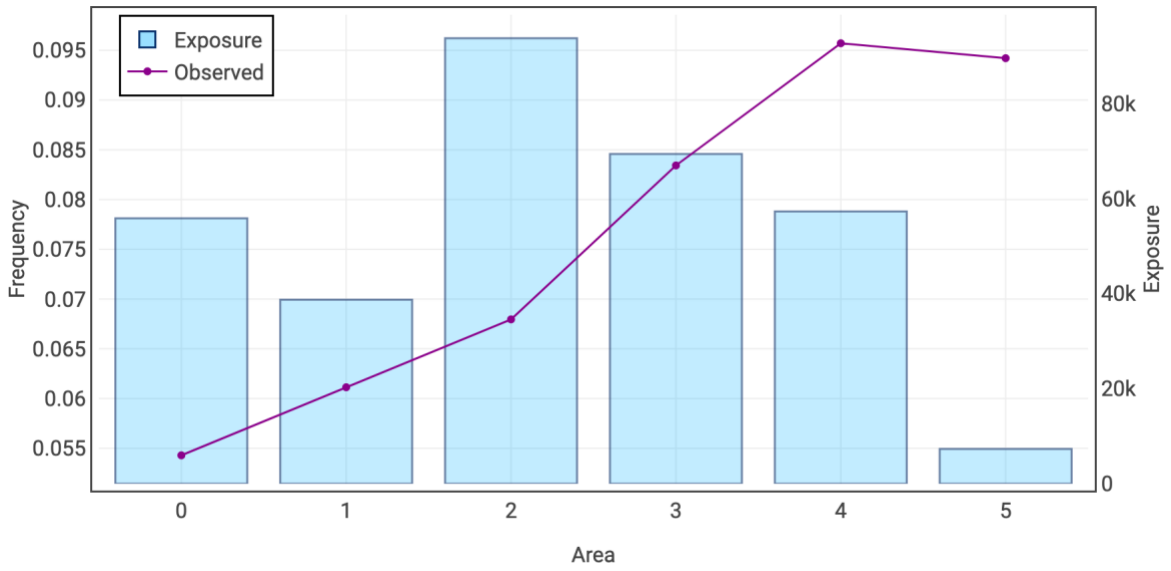



Figure 1.1: The figure represents the observed relationship between the Area variable and the claim frequency. The blue bars represent the sum of the exposures for each individual factor (right y axis) and the purple line represents the aggregated average frequency (total ClaimNb divided by total Exposure).

Intuitively, a linear association may be an appropriate description of the relationship between Area and claim frequency.

Generalized Linear Models provide the standard methodology to compute and visualize such relationships.

A Generalized Linear Model consists of four elements:

1. A target variable (often called response variable), $Y$, which is a random variable whose outcomes are independent with one another.
2. A probability distribution from the exponential family. Equivalently, such a distribution can be defined by a selected variance function and dispersion parameter.
3. A design matrix $X$, a vector of parameters $\beta$ and a linear predictor $\eta = \beta_{\text{intercept}} + X\beta$
4. A monotonic link function $g$, such that $\mathbb{E}(Y) = \mu = g^{-1}(\eta)$.

[3] In the freMtpl dataset, the Area variable takes value from A to E. We mapped these letters to integers for clarity of exposition.

The GLM coefficients $\beta$ are estimated by maximizing the log-likelihood, or, equivalently, by minimizing the *negative* log-likelihood (NLL)

$$\begin{aligned}\hat{\beta} &= \underset{\beta}{\operatorname{argmax}}\{\log(\text{Likelihood}(y, X, \beta))\} \\ &= \underset{\beta}{\operatorname{argmin}}\{-\log(\text{Likelihood}(y, X, \beta))\} \\ &= \underset{\beta}{\operatorname{argmin}}\text{NLL}(y, X, \beta)\end{aligned}$$

To apply such theoretical definition to the Area example, the practitioner will often use statistical softwares such as R. In such language, the definition of a GLM is expressed with the "formula" syntax:

```
model <- glm(formula    = ClaimNb ~ Area,
             offset     = log(Exposure),
             family     = poisson(),
             link       = log,
             data       = learn)
```

The correspondence between the "formula" syntax with the theoretical definition is the following:

1. The **target variable Y** is given by the claim frequency. Such a relationship is implicitly defined by modeling the **claimNb** as the target (left term in the formula argument) and the offset parameter (equal to log(Exposure)).
2. The **distribution** of the target is Poisson as declared by the family argument.
3. The **design matrix** X contains the information defined in the right side of the formula argument. Hence in this simple case $X = X_{\text{area}}$, which is a single column containing the values of the area variable for each observation in the data. The linear predictor $\eta = \beta_{\text{intercept}} + X_{\text{area}}\beta_{\text{area}}$
4. The **link function** which defines the relationship between the linear predictor is the logarithm. The choice of the logarithmic function is convenient as it defines a natural multiplicative structure to the model $\mathbb{E}[Y] = \exp(\beta_{\text{intercept}} + X_{\text{area}}\beta_{\text{area}}) = \exp(\beta_{\text{intercept}}) * \exp(X_{\text{area}}\beta_{\text{area}})$.

The output of the "formula" GLM syntax solution is given by the vector beta = $(\beta_{\text{intercept}}, \beta_{\text{area}})$ which maximizes the likelihood:

| | Parameter |
|---|---|
| Intercept | -2.93339 |
| area_treatment(Area) | 0.140812 |

From a practitioner's perspective, it is preferable to visualize and analyze the GLM output through a **rating table**. The **rating table** utilizes the multiplicative structure given by the logarithmic link:

$$\exp(\beta_{\text{intercept}}) * \exp(X_{\text{area}}\beta_{\text{area}})$$

So that the computation of the predicted frequency is the result of the multiplication of the intercept (also referred to as the base) with other factors, one for each variable. Retrieving the variable factor is an easy table lookup as all **coefficients** (in this case

$$\exp(X_{\text{area}}\beta_{\text{area}})$$

are precomputed on the rating table.

| Variable | Value | Coefficient |
|---|---|---|
| Intercept | | 0.0532 |
| Area | 0 | 1 |
| Area | 1 | 1.1512 |
| Area | 2 | 1.3253 |
| Area | 3 | 1.5257 |
| Area | 4 | 1.7564 |
| Area | 5 | 2.0219 |

The rating table can also be visualized and used as diagnostic for the quality of the model as in the Figure 1.2.

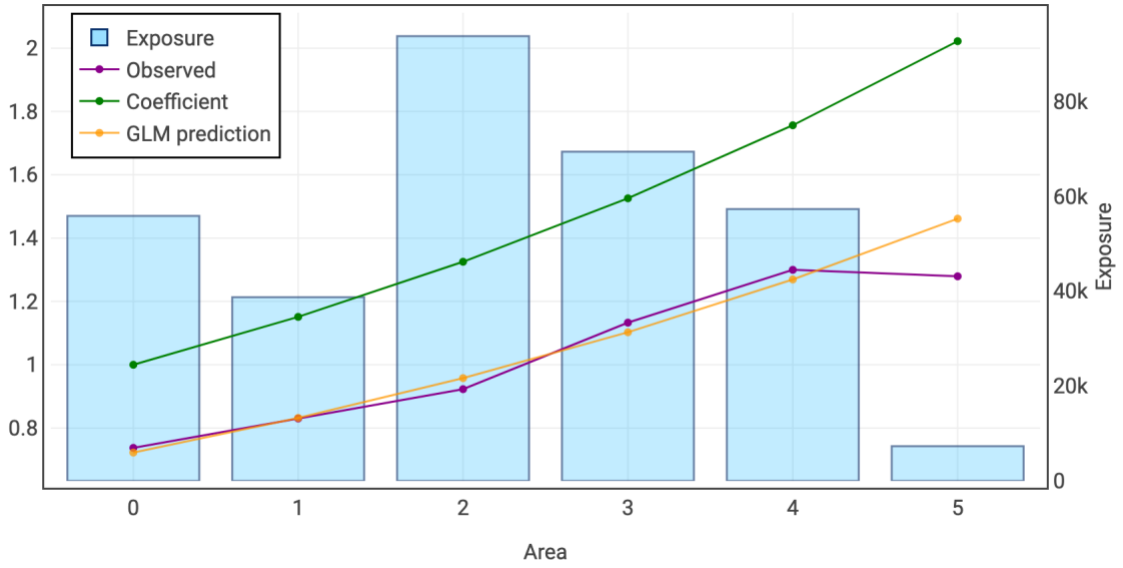
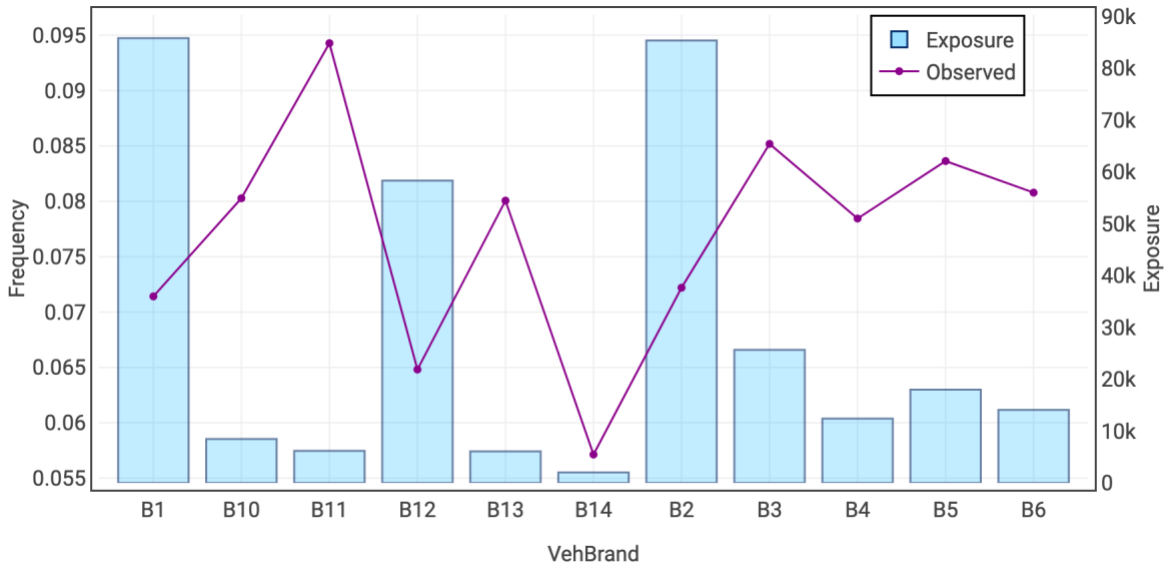



Figure 1.2: The green line represents the values of the coefficients as in the rating table. The orange line represents the (weighted) average of the predictions of the GLM, compared with the Observed and Exposures as defined in Figure 1.1. Observed and GLM prediction have been normalized (both divided by the overall observed frequency) to be compared on the same scale as the coefficient.

A visual analysis shows that the linear (or more precisely, "log" linear) assumption seems appropriate. The GLM predictions follow relatively closely the observed claims for the exception of the area code 5. However, this mismatch is likely explained by the limited amount of exposure of the code as highlighted by the barplot. Expressing a model as a rating table allows the modeler to have a complete understanding and control of the model. This **transparency** (alongside with the versatility of rating tables, described in the sections below) is one of the main factors for which GLMs, presented as rating tables, are the de-facto standard for modeling in insurance.

### The Vehicle Brand variable and categorical variables

Some variables have a relationship with the claim frequency which cannot be described in a continuous fashion, let alone linearly as in the Area example. This is the case for all variables which do not show any implicit ordering. These unordered variables are called **categorical variables**. The variable VehBrand represents the anonymized car brand of the insured vehicle.




Even if the relationship between VehBrand and the frequency cannot be expressed linearly, a "rating table" expression is still possible by associating a multiplicative coefficient to each of the 11 vehicle brands. The natural way to model it via a GLM is by associating one parameter to each one of the values. This is achieved by expressing the vector $\beta = [\beta_{B2}, \beta_{B2}, \dots, \beta_{B14}]$ and applying a dummy encoding (often called "one-hot" encoding) to the design matrix $X$.

| VehBrand |
|---|
| B1 |
| B3 |
| B2 |
| B14 |
| B1 |
| B14 |
| ... |

| VehBrand == B1 | VehBrand == B2 | VehBrand == B3 | ... | VehBrand == B14 |
|---|---|---|---|---|
| 1 | 0 | 0 | ... | 0 |
| 0 | 0 | 1 | ... | 0 |
| 0 | 1 | 0 | ... | 0 |
| 0 | 0 | 0 | ... | 1 |
| 1 | 0 | 0 | ... | 0 |
| 0 | 0 | 0 | ... | 1 |
| ... | ... | ... | ... | ... |

Using the formula language, this encoding action is expressed via the "C" (contrast) syntax

```
formula = ClaimNb ~ C(VehBrand)
```

By default, the solver will fit a parameter for each of the individual brands except for the one with the most exposure. This is a requirement for any GLM solver: if each level of the VehBrand would have its own column, the design matrix will not have maximal rank causing aliasing [Anderson et al. (2007)], which may prohibit the solver from reaching numerical convergence[4]. For the aforementioned reasons, the GLM does not contain a parameter for the Vehicle Brand B1, which is often referred to as the "Base level". This is the case for most modeling packages and softwares.

[4] Rank deficiency of the design matrix generates an infinite number of possible optimal solutions, often leading to divergence behavior of the solver.

Figure 1.3 displays the results of the model, where the base level B1 has a coefficient value equal to 1 (or, equivalently, a parameter beta equal to zero).

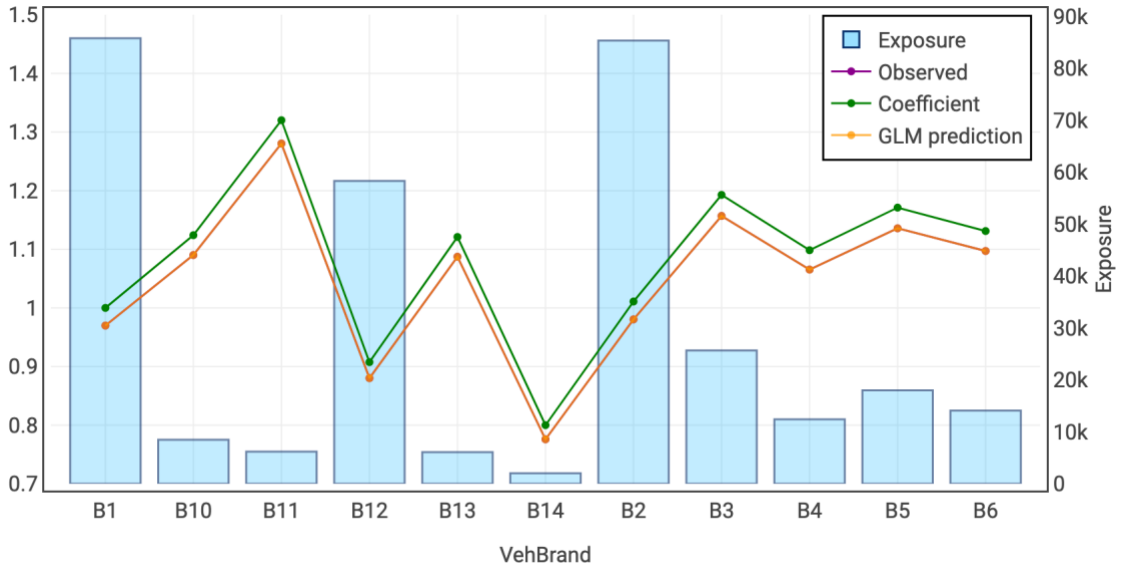



Figure 1.3: Plot of observed claims and univariate GLM predictions for Vehicle Brand (VehBrand)

In the figure, the predicted GLM line is completely overlaid by the observed line. This is not a coincidence: the predicted average on a category will always match the observed average in a Poisson GLM with a single categorical variable. This exact match may not be appropriate for some brands, particularly those with low exposures. Consider as an example: B14 has a relativity of 80% and limited exposure. The modeler may question whether a decrease of risk by 20% is appropriate based on a small amount of data.

P-value analysis provides a tool to the modeler to guide such a decision. Default p-value analysis of solvers will test the hypothesis of whether a given parameter is null. In the VehBrand example, p-values will provide a measure to understand how significantly a given coefficient is different from the modality B1 (which corresponds to the null parameter).

| VehBrand | p-value |
|---|---|
| Intercept / B1 | 0.00000 |
| B10 | 0.00376 |
| B11 | 0.00000 |
| B12 | 0.00000 |

| VehBrand | p-value |
|---|---|
| B13 | 0.01510 |
| B14 | 0.01728 |
| B2 | 0.54602 |
| B3 | 0.00000 |
| B4 | 0.00647 |
| B5 | 0.00000 |
| B6 | 0.00013 |

The de-facto standard on whether to question the "significance" of a specific parameter in insurance and social sciences is 5% [Leahey (2005)]. Based on this rule, the brand B14 is considered to be significantly different from zero since it has a p-value of 1.17 %. The modeler can now support that the brand B14 coefficient should be different from the brand B1. The question on whether the magnitude of the discount of 20% is appropriate cannot be addressed by the GLM.

As per Goldburd et al. (2016) "*for any categorical variable in the model, the estimate of the coefficient for each level is the one which fits the training data best, with no consideration given to the thinness of the data on which it is based*"

The ability to trade off the observed experience with the amount of observed data is a property of Credibility methodologies (see Figure 1.4). Credibility methodologies can determine coefficients that reflect appropriate tradeoffs between the observed data (as result of the GLM fit) and more conservative assumption called complement of credibility, which could be the grand average (almost equivalent to the hypothesis of the parameter B1 to be null) [Casotto, Banterle, and Beraud-Sudreau (2020)].

Unfortunately, because of the properties described above, GLMs are incapable of blending credibility assumptions in their estimates. While a credibility methodology may be more appropriate than a GLM for this univariate analysis, standard credibility methodologies (Classical, Buhlmann) cannot be directly applied to multivariate analysis. This limit of GLMs, and techniques to overcome it will be treated more thoroughly in Chapter 2.

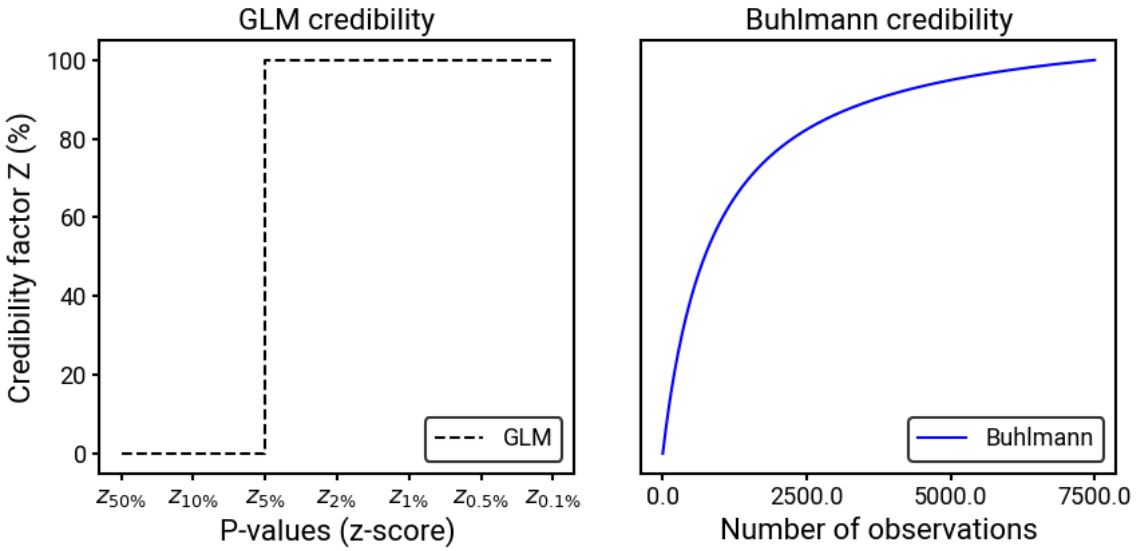



Figure 1.4: Schematic representation of the (absolute) value of the B14 parameter in function of the exposure of the segment, assuming observed average fixed. Buhlmann credibility (on the right) is able to interpolate the coefficient from the complement of credibility (y axis = 0) when exposure is lacking to the GLM coefficient. Lack of credibility of GLMs means that only two modeling choices are possible, whether to exclude (p-value above 5%) or include the full experience.

## The Vehicle Age variable and binning of ordinal variables

The Vehicle Age (VehAge) variable represents the age of the insured cars (in year). Figure 1.5 provides the univariate effect of the variable.

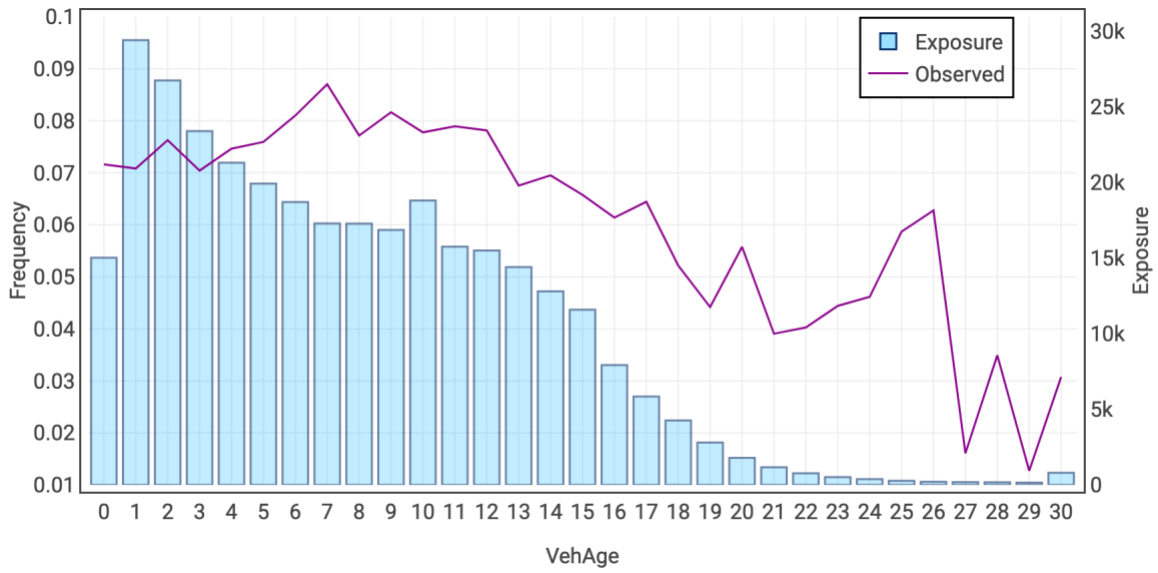
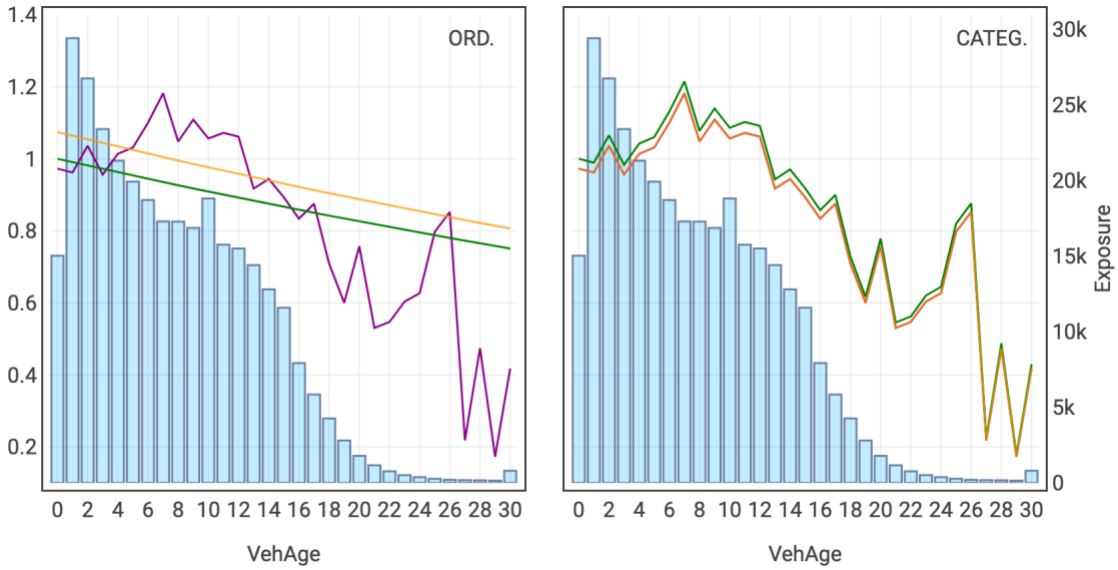



Figure 1.5: Plot of observed claims for vehicle Age variable (VehAge)

Although vehicle age is an ordinal variable, assuming a linear association between vehicle age and frequency doesn't properly represent the underlying behavior. A linear trend fails to capture the slight risk increase from ages 6 to 12, followed by a sharp risk decrease. On the other hand, fitting the variable categorically, as with VehBrand, is not suitable, as the resulting modeled relationship may follow wiggles observed in the data too closely, often driven by random noise.




Result of an univariate GLM treating the variable vehicle age as an ordinal variable (left) and as a categorical variable (right). On the left, we can see the nonlinear relationship between the frequency and the vehicle age that cannot be represented by a linear model. On the right, the model obtained by considering the vehicle age as categorical would capture the noise in the data and would be prone to overfitting.

It's common to observe nonlinearity, such as with Vehicle Age and frequency, in insurance data. To establish a more appropriate relationship between this variable and the target, manual feature engineering is often necessary to ensure visually explainable coefficient trends. Chapter 5 of the CAS Monograph on GLMs by Goldburd et al. (2016) delves into these strategies in detail. We will present two strategies and, in this section, start with the technique of binning (also known as levels grouping) for the Vehicle Age variable. This involves grouping ages into buckets of consecutive values and fitting those buckets as categorical variables. The rationale is that creating categories with larger exposures mitigates against the "wiggling" effect caused by independently fitting each level. Defining the boundaries of these buckets requires the practitioner to guess-estimate them based on visual trend analysis. For instance, a modeler might opt to define three buckets with cut points at ages 6 and 12, as depicted in Figure 1.6, which illustrates the resulting GLM. In general, the number of buckets created and their edges will be selected by the modeler by finding the right balance between:

- **The exposure** (or significance of the effect) of each bucket: buckets with insufficient exposure will lead to unreliable models (the variance of the model created will be too high)
- **The granularity** of the model created: if the number of buckets created is too low, the models created will be too coarse (the bias of the model created will be too high)

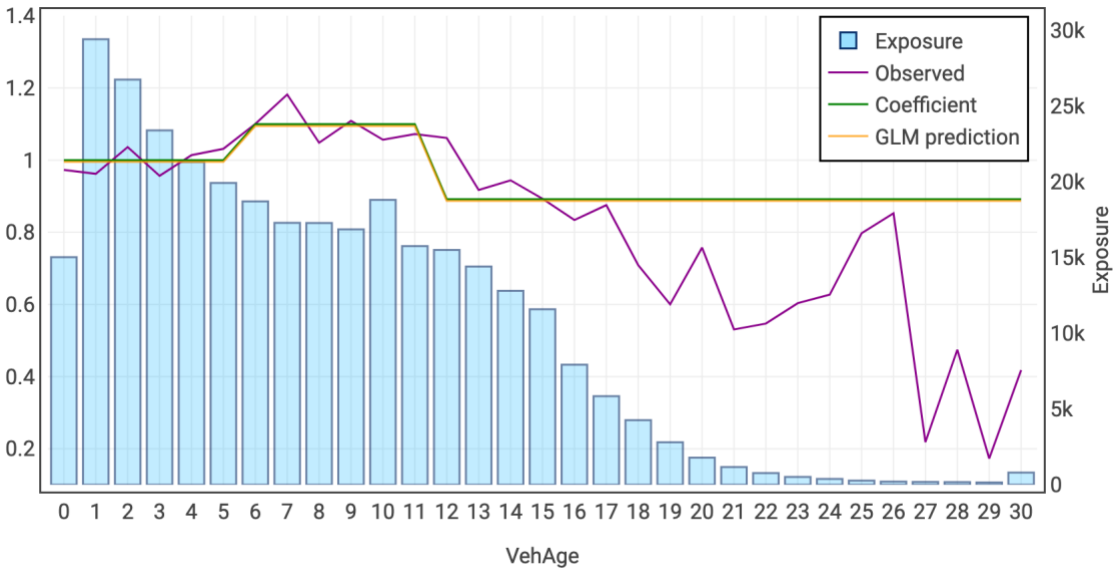



Figure 1.6: Ungrouped display of GLM coefficient of VehAge with cut points at 6 and 12.

Similarly to VehBrand, the modeler should also assess whether the jump between ages 6 to 12 is appropriate. A p-value analysis indicates that both the central bin (ages 6-12) and the right bin (ages 12+) are significantly different from the base level bin (ages 0-6).

Based on this univariate analysis, the modeler may consider adding a supplementary cut point at age 20. However, increasing the number of bins for an ordinal variable in GLM has two potential downsides. Citing Goldburd et al. (2016):

- **Continuity in the estimates is not guaranteed:** Allowing each interval to move freely may not always be a good thing. The ordinal property of the levels of the binned variable have no meaning in the GLM; there is no way to force the GLM to have the estimates behave in any continuous fashion, and each estimate is derived independently of the others. Therefore, there is a risk that some estimates will be inconsistent with others due to random noise. […]
- **Variation within intervals is ignored:** Since each bin is assigned a single estimate, the model ignores any variation in severity that may exist within the bins. […] Of course, we could refine the interval boundaries to split that bin into two or more smaller ones. Doing so, however, would thin out the data, reducing the credibility of the estimates, thereby making them more susceptible to noise.

In Chapter 3 of this paper, we will delve into these important considerations and propose a solution to overcome the limitations of the grouping approach. By changing the assumptions behind ordinal variables, we will be able to incorporate level grouping into the inherent structure of penalized regression.

### The Driver Age variable and inclusion of nonlinear terms

The variable Driver Age (DrivAge) represents the driver's age in years. As depicted in Figure 1.7, there exists a nonlinear association between driver age and claim frequency.

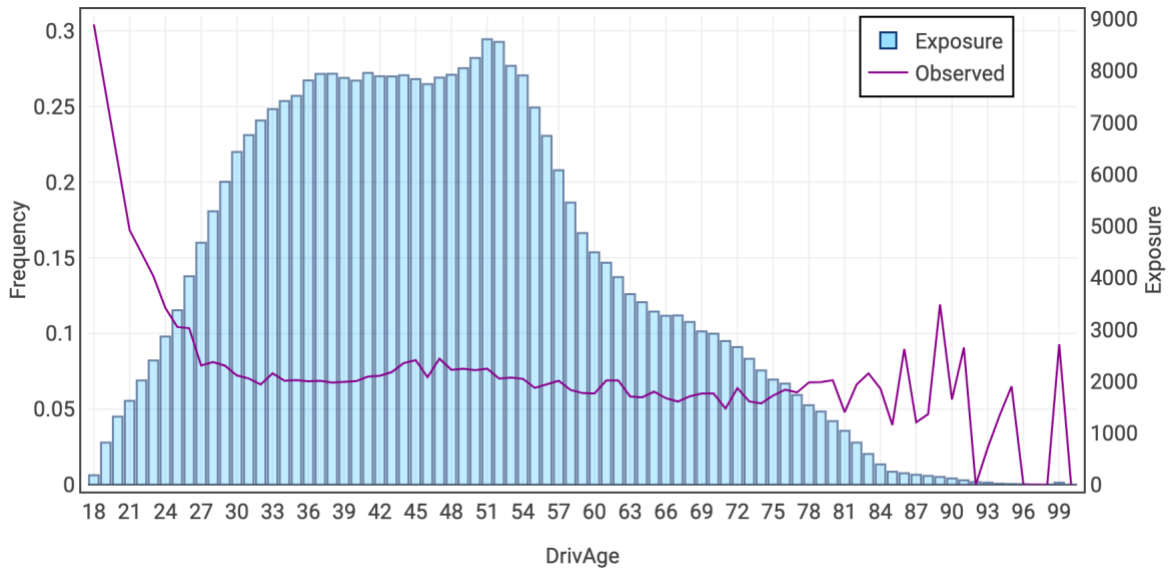



Figure 1.7: Plot of observed claims for driver age variable (DrivAge)

Describing the nonlinearity of Driver Age is challenging, as it appears linear from age 18 to 24 but then warps toward a U-parabolic shape from 24 to 42 before changing concavity again. To model this nonlinearity, a variety of functions such as polynomials and piecewise linear functions can be selected as parameters of the GLM. These parameters transform the data to make it appear linear to the GLM solver. However, selecting the appropriate function can be difficult without overfitting. The CAS monograph by Goldburd et al. (2016) provides guidance on how to make modeling decisions while avoiding over-complex shapes. To be concise, we will use the DrivAge parametrization defined in Wüthrich and Merz (2023) without further discussing the modeling decisions and transformations involved.

```
ClaimNb ~ DrivAge + I(DrivAge**2) + I(DrivAge**3) + I(DrivAge**4) +
log(DrivAge)
```

Figure 1.8 represents the results of the fit.

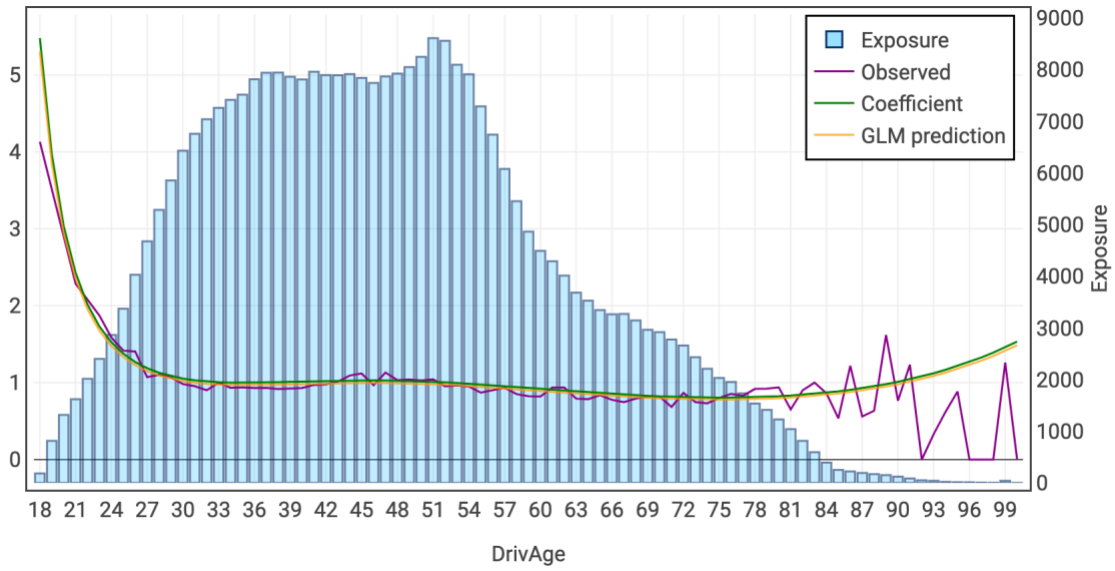



Figure 1.8: Plot of the coefficient according to the formula above. Coefficients were normalized so to put mode (Age = 36) to 1.0.

The mixed 4th degree polynomial parametrization chosen for Driver Age improves the model fit compared to a linear model. A p-value analysis indicates that all five parameters are unlikely to be null. The modeler may rationalize the introduction of the logarithm to account for the steep decrease in risk for very young ages, and the use of a fourth degree polynomial to interpolate a function with two local minima around ages 30 and 70. Despite these arguments, it remains challenging to fully justify the choice of these specific nonlinear terms from among an infinite number of similarly-behaving functions. In general, variable transformations such as polynomials and hinges aim to describe nonlinear relationships by combining known functions. However, the choice of function can vary greatly among modelers, potentially leading to different final parametrizations. Additionally, combining complicated variable transformations can affect the stability of the model. For instance, polynomial encoding can behave erratically at the edges of the data, particularly for higher-order polynomials Goldburd et al. (2016), as shown in Figure 1.9. The removal of the logarithmic term to the coefficient of Driver Age causes an unexpected abrupt increase in trends on the right-hand side, degrading the model fit. This “whip” effect can be explained by the inherent complexity of the 4th degree polynomial.

This example shows the power of variable transformations in capturing nonlinear effects. However, it also demonstrates the expertise, effort and subjectivity involved in crafting the appropriate transformations for numeric variables to efficiently represent their effects in a highly iterative procedure.

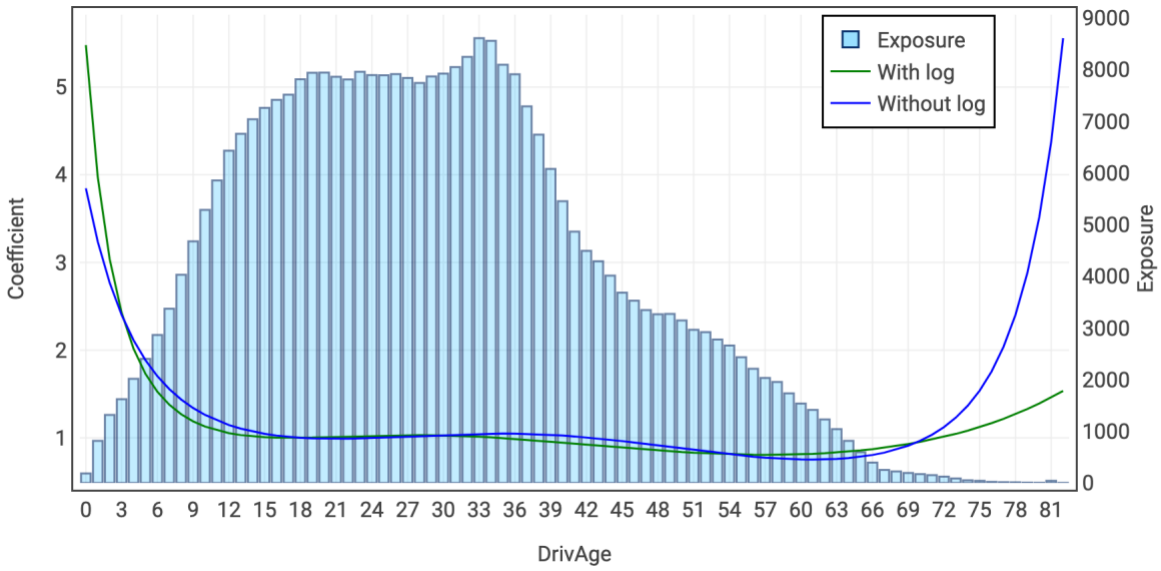



Figure 1.9: The green line represents the driver age as in the GLM formula ClaimNb ~ DrivAge + I(DrivAge**2) + I(DrivAge**3) + I(DrivAge**4) + log(DrivAge); the blue line represents the driver age without the log term, i.e., ClaimNb ~ DrivAge + I(DrivAge**2) + I(DrivAge**3) + I(DrivAge**4).

## 1.3 Multivariate analysis

In our analysis of the Fremtpl data, we presented the key techniques used to build GLMs with insurance data on univariate modeling examples. However, it is important to note that correlations between policy attributes can have a significant impact on the resulting model. In this section, we will illustrate how a multivariate model differs from the individual components built in the univariate analysis. Specifically, some modeling decisions that were validated on an univariate basis may not hold when considering other dependencies and correlations. To explore this further, we will use the multivariate model from Schelldorfer and Wuthrich (2019), which is also referred to as the “GLM2” model in Wüthrich and Merz (2023) , whose formula is:

```
ClaimNb ~ C(Region) + C(VehBrand) + C(VehGas) +
                      Area + BonusMalusGLM +
                      C(VehPowerGLM) + C(VehAgeGLM) +
                      DrivAge + log(DrivAge) + I(DrivAge^2) +
                      I(DrivAge^3) + I(DrivAge^4)
```

This model applies the similar transformations used in the univariate analysis on the four relevant variables. Furthermore, it includes five additional policy attributes which may have been transformed, as flagged by the postfix “GLM”.

## The Area variable

We begin our analysis by examining the variable Area, which appears in both the univariate and multivariate models as a (log) linear effect. Figure 1.10 displays the fit quality and compares the coefficients obtained from the univariate and multivariate models.

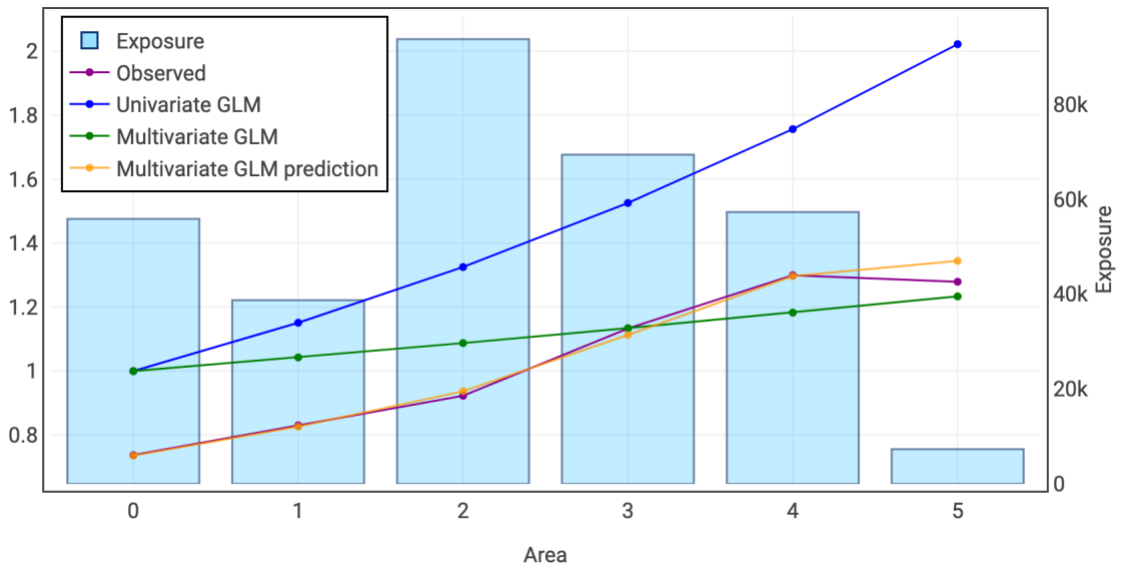



Figure 1.10: Comparison of the univariate GLM estimates (blue) on the Area variable with the multivariate estimates. Multivariate GLM prediction (orange) represents the predictions of the multivariate model.

The linearity assumption of the Area variable seems to hold even in the multivariate model. However, we note that the individual impact of the Area's coefficients in the multivariate model is smaller compared to that of the univariate analysis. This can be attributed to the correlation of Area with the Density variable, which represents the density of inhabitants in the city of the driver's residence. Both logically and quantitatively (as demonstrated by a correlation analysis), the observed effects can be explained by this consideration.

The parameter for the slope of the linear trend for Area yields a p-value of 3.6%, which remains below the standard 5% threshold. Hence, retaining of this parameter can be justified. However, the modeler may still have doubts about its inclusion and may want to perform additional analyses outside the scope of this paper.

## The Vehicle Brand variable

The correlation structure has a significant impact on the VehBrand variable. Figure 1.11 compares the results of the multivariate and univariate models.

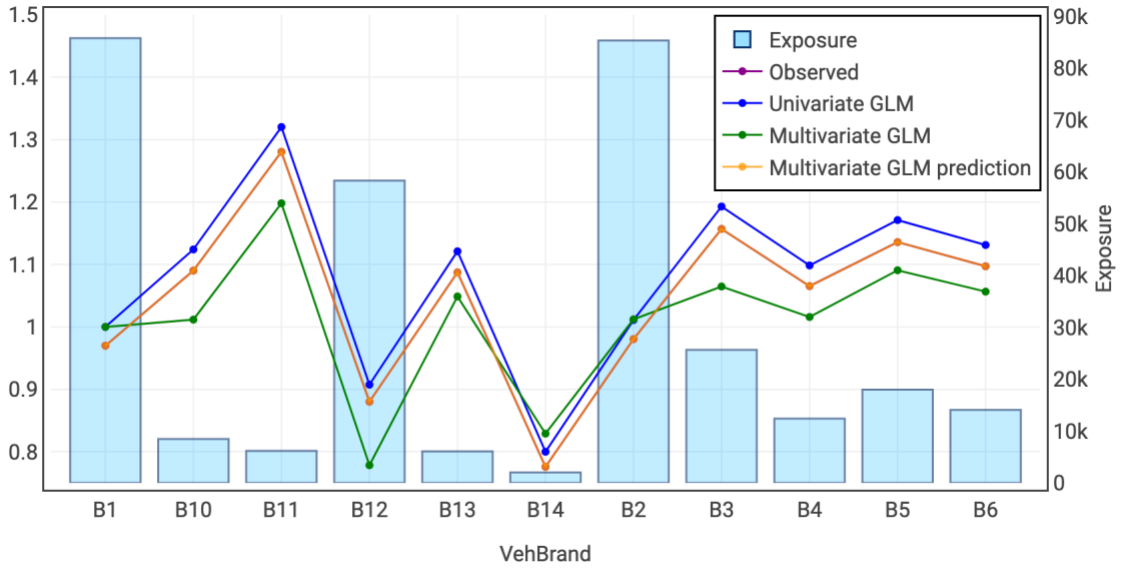



Figure 1.11: Comparison of the univariate GLM estimates (blue) on the Vehicle Brand variable with the multivariate estimates. Multivariate GLM prediction (orange) represents the predictions of the multivariate model.

Both models predict the data accurately, as expected from a GLM. However, the overall impact of the VehBrand coefficients is reduced in the multivariate model due to the correlation effects with the other variables in the GLM.

Comparing the p-values from multivariate analysis to those from the univariate analysis, we observe that the p-values for vehicle brands B10 and B11 shift from well below the 5% threshold to well above it, signifying that their coefficients are no longer statistically significant. Notably, the p-value for vehicle brand B14 changes from well below to slightly below the threshold. Consequently, as in the univariate analysis, the coefficient for brand B14 in the multivariate analysis raises questions for the modeler. Once correlations are included in the modeling process, the GLM indicates a risk discount of -17.3% for level B14. The p-value analysis for this level, at 4.6%, is close to the 5% threshold, raising questions about whether this level justifies such a substantial decrease in risk.

| | GLM Coefficient | p-value |
|---|---|---|
| B1 | 1.00000 | nan |
| B10 | 1.01167 | 0.78277 |
| B11 | 1.19804 | 0.00005 |
| B12 | 0.77841 | 0.00000 |

| | GLM Coefficient | p-value |
|---|---|---|
| B13 | 1.04876 | 0.31601 |
| B14 | 0.82906 | 0.04617 |
| B2 | 1.01219 | 0.50507 |
| B3 | 1.06474 | 0.01297 |
| B4 | 1.01589 | 0.65058 |
| B5 | 1.09073 | 0.00289 |
| B6 | 1.05653 | 0.09113 |

To resolve this dilemma, the practitioner can either include or exclude the B14 parameter based on a binary decision or apply a post-modeling selection using an ad-hoc credibility adjustment to the VehBrand coefficient. However, as known credibility procedures work only on an univariate setting, this "post-modeling" approach may not be appropriate due to the correlation between variables.

This gets back to the old issue that a sequence of steps, each optimal individually, may not be optimal in the aggregate [Klinker (2010)].

### The Vehicle Age variable

We now turn our attention to the VehAge variable and its impact on the model's structure. Figure 1.12 displays the fit quality of the variable and compares the coefficients from the univariate analysis.

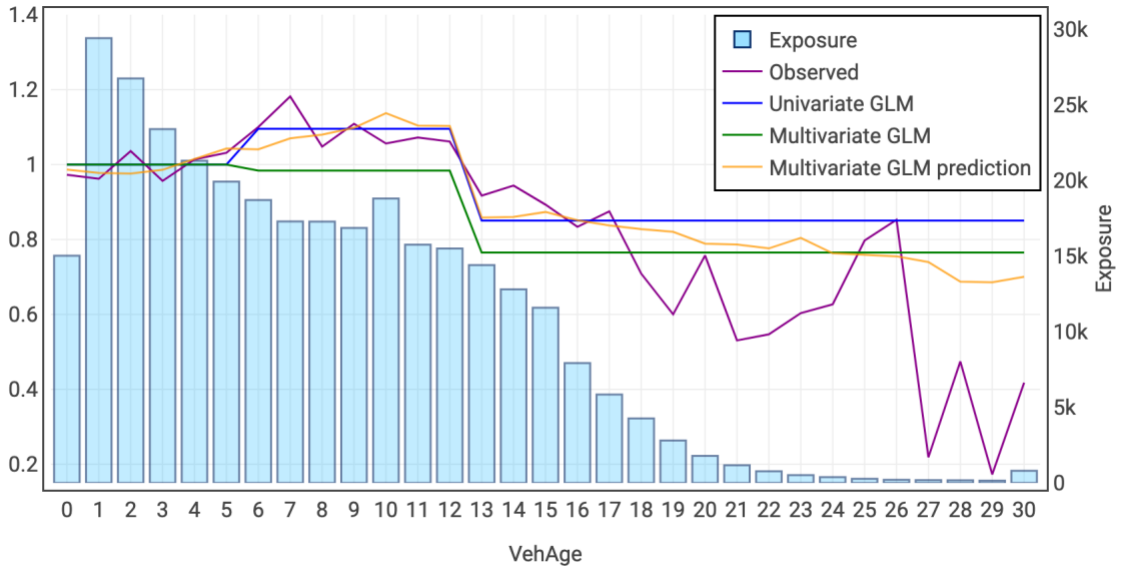



Figure 1.12: Comparison of the univariate GLM estimates (blue) on the Vehicle Age variable with the multivariate estimates. Multivariate GLM prediction (orange) represents the predictions of the multivariate model.

The inverted U-shaped trend of the VehAge coefficients in the univariate analysis changes to a more decreasing trend in the multivariate model. Interestingly, the difference between the two age groups (0-6 and 6-12 years) is now minimal. A p-value analysis indicates that the GLM parameter corresponding to the central bin is not significantly different from the null hypothesis (p-value = 29%). Since the base level refers to ages from 0 to 5 and the null hypothesis is $\beta_{Age=[6-12)} = 0$, we can conclude that the refined grouping is not significant.

| VehAge | p-value |
|---|---|
| 6-12 | 0.29007 |
| 12+ | 0.00000 |

Following the parsimony principle, the modeler may choose not to model the group 6-12 as an independent parameter, effectively giving the same coefficient for all ages 0-12. This example demonstrates that modeling decisions that were sound in one step of the model building can be invalidated by changing the variables in the model. Each iteration in GLM building, whether by adding a new variable or different parametrization of a previously included factor (binning, polynomial, etc.), may invalidate or reduce the validity of previous modeling decisions. This means that the modeler needs to continuously reexamine every decision made in previous iterations. This time-consuming procedure will be detailed in depth in Chapter 3, which presents a more robust approach to obtain statistically sound data-driven binning.

### The Driver Age variable

We conclude the analysis of the multivariate model by comparing in Figure 1.13 the multivariate driver age fit with the univariate fit. As for other variables, the individual contribution of the driver age variable is reduced. The captured trend still shows a steep decrease in risk as age increases from 18 to 30, followed by an increase of the risk. However, the risk trend for old drivers above 80 years is inverted, most probably due to the limitations of modeling this variable via polynomial transformations.

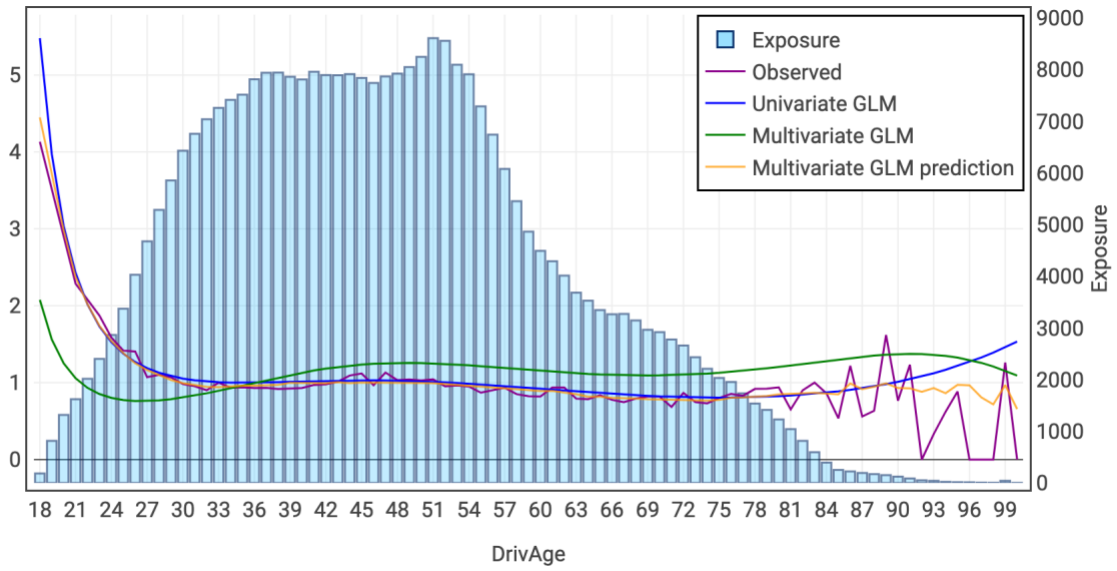



Figure 1.13: Comparison of the univariate GLM estimates (blue) on the Driver Age variable with the multivariate estimates. Multivariate GLM prediction (orange) represents the predictions of the multivariate model.

## 1.4 Conclusion

This chapter has introduced some of the most commonly used techniques in standard GLM modeling and has highlighted the challenges that modelers face when constructing predictive models with GLMs. Through a detailed analysis of real-world insurance data, we have underscored two key points:

1. Possible drawbacks of the GLM methodology, and the strategies and iterations that the practitioners use to mitigate or solve these shortcomings, as summarized in the table below;
2. How adding or transforming variables in the model may invalidate previous modeling choices due to the changed correlation structure, thus increasing the need for extensive checks at each modeling iteration.

Modelers should pay close attention to each modeling step and thoughtfully evaluate the influence of each variable and modeling decision on the overall model. In the following

chapters, we present a more robust approach to these two matters, which addresses the challenges highlighted in this chapter. By changing the assumptions underlying the behavior of ordinal variables, we can avoid these pitfalls and handle nonlinearities in a multivariate and universal approach.

| Example variable | Transformation applied | Limitation of transformation |
|---|---|---|
| Area | None | Applies to simple ordinal relationships |
| VehBrand | Categorical encoding | Lack of credibility procedure for levels with low exposures |
| VehAge | Binning | Continuity in the estimates is not guaranteed. Variation within intervals is ignored. |
| DrivAge | Nonlinear / Polynomial transformation | Result of several iterations. Unstable when performing extrapolation. |

# 2. Credibility and penalized regression

The previous chapter emphasized the substantial mastery and skills required to construct GLMs in order to capture nonlinear effects and manage low-exposure, low-credibility segments. In a multivariate model, modeling each variable involves multiple manual iterations, with each iteration potentially invalidating prior modeling decisions made on other variables. As the number of analyzed variables increases, the number of potential dependencies grows exponentially, making it a labor-intensive process to strike the right balance among these moving pieces. Many of these manual iterations are necessary to overcome two main limitations of GLMs, as noted in Goldburd et al. (2016):

- **Credibility:** Full credibility is given to the data for each coefficient, with no regard to the thinness on which it is based;
- **Nonlinearity:** Predictions must be based on a linear function of the predictors.

This chapter will explore how penalized regression provides an efficient solution to blend **credibility** assumptions into GLMs, both in theory and in practice. Here, we will focus on the application of penalized regression to multivariate GLM modeling for categorical variables; detailed discussion for ordinal variables will be presented in Chapter 3. This chapter will summarize previous research on the connection between credibility and penalized regression, including the work by Casotto, Banterle, and Beraud-Sudreau (2020) and by Miller (2015). For a more in-depth discussion of these topics and the majority of the proofs, we refer to those papers.

## 2.1 Credibility and univariate GLMs

*Credibility, simply put, is the weighting together of different estimates to come up with a combined estimate.*

– Foundations of Casualty Actuarial Science [Casualty Actuarial Society, 2001]

When observed data is limited, credibility provides a reliable statistical framework for developing robust risk estimates. As seen in Chapter 1, the VehBrand variable highlights concerns about the amount of underlying data required to develop estimates. The univariate GLM estimate suggests that Brand B14 should receive an approximately 20% discount in risk frequency. While the direction of the estimated discounts and its bounds (between 0 and 20%) is evident, the total magnitude may not be justifiable. Unfortunately, GLMs do not provide a satisfactory answer to these concerns. In this case, practitioners are left with the following three options:

1. Maintain a single coefficient assignment for B14, effectively accepting the 20% discount.

2. Ignore the B14 segment by absorbing the B14 coefficient into the base level, implying no discount is applied and $\beta_{\text{B14}} = 0$. This approach can lead to a questionable solution due to potential dissimilarities between the two levels.

3. Group the coefficient for B14 with those of other levels.

However, the most appropriate answer may lie in between the first two scenarios, with an estimate between 0 and -20%. Credibility provides a framework to decide the weight of this tradeoff, denoted by the credibility factor $Z$. The formula for the estimate is:

$$\text{Estimate} = Z \times \text{Observed Experience} + (1 - Z) \times \text{Complement of Credibility},$$

where the complement of credibility is an alternative estimate of the risk. For example, if we consider the complement of credibility to be the grand average of frequency in the database, one can transpose the formula to the GLM framework with similar arguments as in Casotto, Banterle, and Beraud-Sudreau (2020), with

$$\text{B14 Coefficient}_{\text{Credibility}} = Z_{B14} \times \text{B14 Coefficient}_{\text{GLM}}.$$

In the equations above, $Z$ is a factor between 0% and 100% and always increases with the number of observations. When the credibility factor $Z$ equals 100%, the estimate is equivalent to the observed experience, which is the same as GLM. Therefore, it can be concluded that GLM provides 100% credibility to the data.

The explicit credibility formula in Werner and Modlin (2010) can be applied only to a univariate setting. Blending credibility with a GLM requires applying modifications to the GLM framework, such as by applying penalized regression.

## 2.2 Penalized regression

The intuition behind penalized regression is simple. GLMs always maximize the likelihood, which essentially assigns 100% credibility to the data, regardless of the underlying exposure. To overcome this drawback, a "penalty" term can be added to the GLM optimization process, which allows for *simultaneous* optimization of a tradeoff between pure goodness of fit (which GLMs optimize exclusively) and a desirable structure of the coefficients prescribed by the penalty. For example, adding this additional term may turn the optimization in favor of a model with a desirable property of coefficient shrinkage in cases where the credibility of a segment is limited.

The general formula of penalized regression is

$$\hat{\beta} = \underset{\beta}{\text{argmin}} \text{NLL}(y, X, \beta) + \lambda \text{Penalty}(\beta), \qquad (2.1)$$

where $\lambda \geq 0$ is a positive number, called the **penalty parameter** . The penalty parameter $\lambda$ acts like a dial, allowing us to adjust the balance between the goodness of fit to the training

database and the a-priori structure of the coefficients. By tuning $\lambda$, we can assign more or less importance to each of these aspects.

There are two main, well-established types of penalties[5]:

- The **ridge** penalty is given by the sum of squares (or $l2$ norm squared) of $\beta$, i.e., $\sum_{j=1}^{p} \beta_j^2$.
- The **lasso** penalty is given by the sum of absolute values (or $l1$ norm) of $\beta$, i.e., $\sum_{j=1}^{p} |\beta_j|$.

Penalized regression techniques such as lasso and ridge are, at their core, very similar to GLMs in their mathematical specification [Goldburd et al. (2016)]. In fact, both penalized GLMs and simple GLMs share the same model or hypothesis structure:

- The estimated parameter is related to a **linear combination** of the explanatory variables by a **link function**.
- Observations are assumed to follow a **distribution** around the estimated parameter.
- The **table-based output** and the variable parametrization are preserved.

The primary difference is given by the choice of the penalty (lasso or ridge) and the value of the penalty parameter. These choices will determine the difference between the estimates of the parameters beta of a penalized model compared to the unpenalized (simple GLMs).

The choice of the penalty and the $\lambda$ parameter are, in fact, equivalent to the addition of statistical hypotheses. In lasso/ridge (but not in simple GLMs), the coefficients of the linear combination are assumed to follow a predefined distribution; in simple GLMs, there is no such assumption. The strength of such an assumption is proportional to the penalty parameter $\lambda$. When $\lambda$ is zero, penalized regression is exactly a GLM.

This statistical interpretation of the penalty is a powerful tool that highlights the connection between Buhlmann credibility and penalized regression.

## 2.3 The equivalence between ridge regression and Buhlmann credibility

Bayesian modeling provides a powerful paradigm which makes the connections (and in some cases, equivalencies) between credibility and penalized regression. The gist is simple: Buhlmann credibility is a Bayesian model, and so are any solution to penalized regression. When the underlying Bayesian model is the same, we can deduce that Buhlmann credibility is equivalent to penalized regression.

In a Bayesian model, on top of the standard hypothesis applied to the observations (as in a GLM), a distributional assumption is made on the coefficients of the model themselves (called

[5] In some applications, these two penalties are used simultaneously. The resulting combination is referred to as the "Elastic Net" penalty, whose analysis is outside of the scope of the paper.

the prior distribution). This assumption describes our a priori knowledge and uncertainty over the value of the coefficients (hence the name), which are now random variables, and allows for the inclusion of some additional structure on the coefficients' estimates.

Figure 2.1 illustrates an example of a Bayesian model which combines:

1. The statistical assumption over the observation is that we observe outcomes which are normally distributed around the parameters of the model (as in a linear regression or Gaussian GLM).
2. The additional assumption on the parameters of the model, called the **prior** assumption. Here the **prior** is a normal distribution with zero mean.

Implicitly, we expect that parameters which are large in magnitude to be extremely unlikely or, at least, less probable than parameters close to the mean (in this case, zero). In general, the choice of the prior distribution allows us to formalize common sense intuitions in a robust mathematical framework.

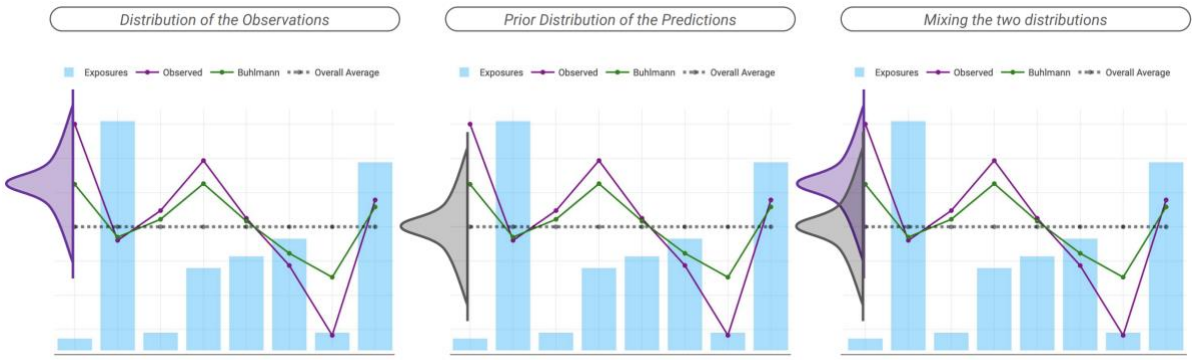



Figure 2.1: Statistical hypotheses behind the ridge regression and Buhlmann credibility. Left panel represents the distribution of the observations, assumed to be a normal distribution (purple) centered at the model estimates (green). Middle panel shows the Buhlmann credibility prior hypothesis: the coefficients are a-priori assumed to follow a normal distribution centered (gray) at zero (the complement of the credibility). Right panel displays the combination of these two probability distributions, which changes the likelihood of the overall model.

In univariate modeling, the estimates of this model are equivalent to a Buhlmann credibility model with a Gaussian assumption on the observations, as proven in the Bayesian equivalence of Buhlmann by Jewell (1974). Furthermore, the parameter $K$ in the Buhlmann formula is given by the ratio of the variances of the two normal distributions $K = \frac{\tau^2}{\sigma^2}$. We can also estimate the same model by maximizing the log-likelihood (as in GLMs).

As shown in Casotto, Banterle, and Beraud-Sudreau (2020), the solution can be computed by solving

$$\beta = \underset{\beta}{\operatorname{argmin}} \mathrm{NLL}(y, X, \beta) - \log\left(p_{\text{prior}}(\beta)\right)$$

Where the left term is the negative likelihood of the assumption over the observations, and the right term the log-likelihood of the prior distribution.

Under the hypothesis in Figure 2.1, the formula simplifies further to the ridge regression:

$$\beta = \underset{\beta}{\operatorname{argmin}} \frac{1}{2}\sum_{i=1}^{n}(y_i - X_i\beta)^2 + \frac{\lambda}{2}\sum_{j=1}^{p}\beta_j^2$$

Where the penalty parameter is $\lambda = K = \frac{\tau^2}{\sigma^2}$.

### Role of the penalty parameter in ridge or Buhlmann credibility

The Buhlmann credibility weight $Z$ is computed via the formula $Z = \frac{n}{n+K}$, with $K$ being a quantity estimated from the data and $n$ the underlying number of observations. For a fixed number of observations, the smaller the $K$, the higher the credibility towards the data. When $K$ is equal to zero, then the data is 100% credible, as with GLMs. In an univariate normal setting, the penalty parameter $\lambda$ is equal to the $K$ parameter as described above. In a multivariate setting, the relationship between the $\lambda$ parameter and the credibility given to the data still holds true.

We can illustrate this fact via Figure 2.2, which shows how the estimate changes for VehBrand for three different values of ridge penalty parameters.

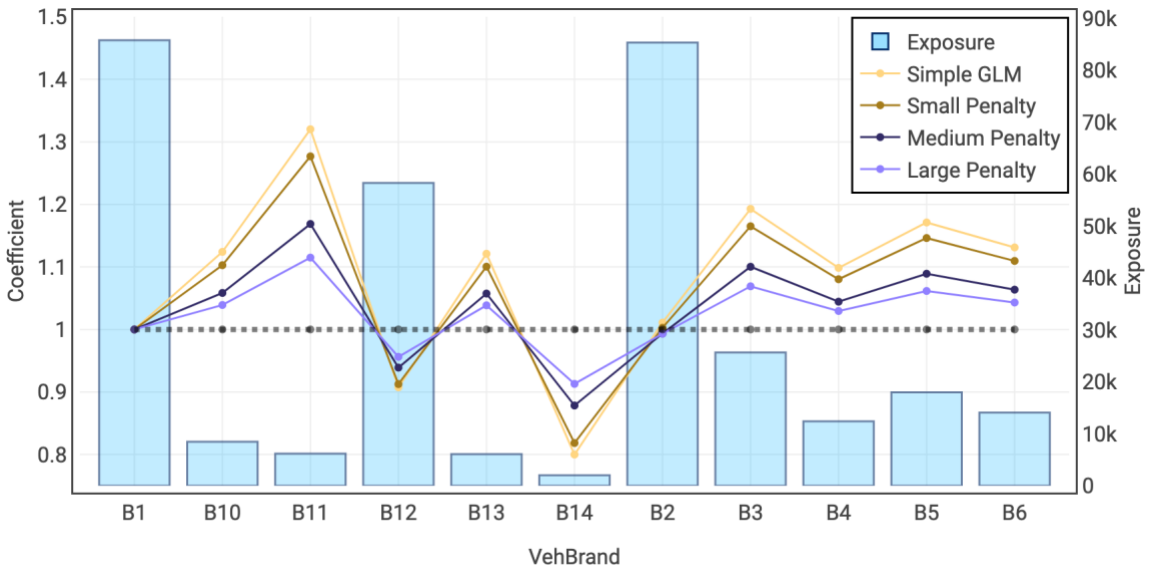



Figure 2.2: Ridge fit for various penalty values $\lambda$ on the univariate Vehicle Brand model ClaimNb ~ C(VehBrand). Applying Buhlmann credibility with different parameter values $K$ would lead to the same results. Simple GLM represents the unpenalized GLM estimates (which give 100% credibility to the data). The black dashed line at 1.0 represents the complement of the credibility, or equivalently, the center of the prior hypothesis of ridge penalty.

The magnitude of the penalty parameter $\lambda$ can be summarized as

- **Small** $\lambda$ (small penalty):
  - Coefficients are **close** to the simple, unpenalized GLM (actually equal to the GLM when $\lambda = 0$).
  - The **prior** hypothesis (coefficient are centered around zero) has **weak** influence on the model.
- **Medium** $\lambda$ (medium penalty):
  - Coefficients are in between the simple, unpenalized GLM and the complement of credibility (in this case, the black dashed line centered at 1.0)
  - There is balance between the influence of the **prior** hypothesis and the observed data.
- **Large** $\lambda$ (large penalty):
  - Coefficients are close to the complement of credibility.
  - The **prior** hypothesis has **strong** influence on the coefficients.

The optimal penalty parameter value $\lambda$ varies depending on the database and the scope of the modeling. It is a common practice to choose the $\lambda$ value using cross validation [Hastie et al. (2009)]. We refer to Chapter 4 for a more detailed explanation of best practices to decide the best penalty parameter value.

## 2.4 Lasso regression and parameter selection

Ridge regression incorporates Buhlmann credibility in a GLM by allowing the model to continuously interpolate the estimates from the grand average to the observed data. Regardless of the penalty value $\lambda$, ridge will not exactly set model parameter values to zero.

This is a significant shift from the GLM procedure, where the modeler will often decide to include or exclude parameters using tools such as p-values.

From this perspective, lasso regression combines elements of simple GLMs and ridge regression. It can set model parameters exactly to zero, similar to level selection in classic GLM modeling. At the same time, it can also shrink coefficients, akin to a credibility approach.

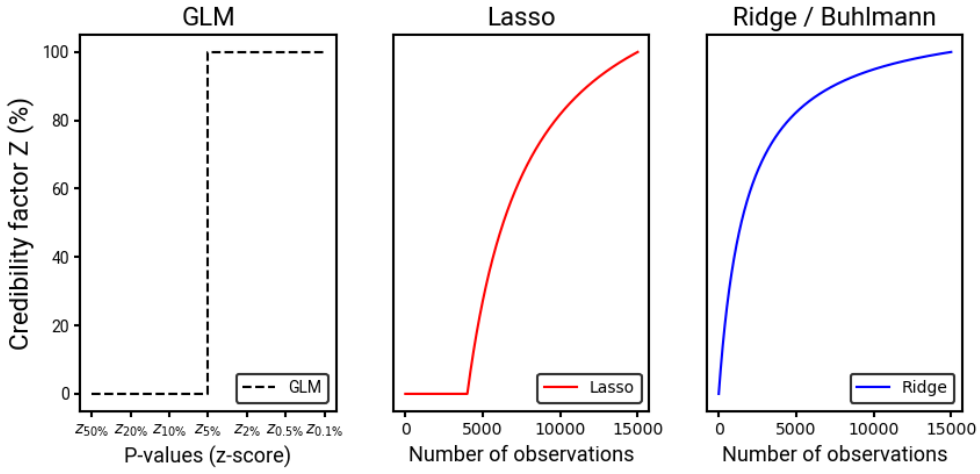



Figure 2.3: Schematic representation of the absolute value of the B14 parameter as a function of segment exposure, assuming a fixed observed average. Right panel: Ridge regression, or equivalently, Buhlmann credibility, has the ability to interpolate the coefficient based on the complement of credibility when exposure is lacking. Left panel: A classic, non-penalized GLM lacks the notion of partial credibility. This results in a binary modeling choice: either exclude the parameter (if the p-value exceeds 5%) and entirely discard the information provided by the data or include it and fully trust the data. Center panel: Lasso regression represents a blend of these two approaches.

This difference can be justified both in theory and in practice.

Lasso penalized regression (like all penalized regression models; see Miller (2015)) derives from a Bayesian model whose **hypothesis on the coefficients (prior)** is a Laplace distribution (see Figure 2.4):

$$\text{Laplace}(\lambda) = \frac{1}{2\lambda}\exp\left(-\frac{|x|}{\lambda}\right)$$

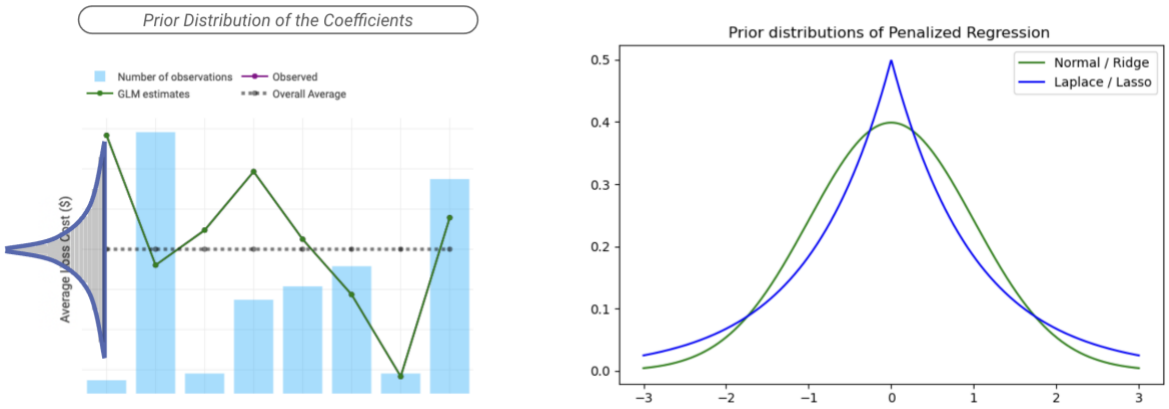



Figure 2.4: The prior hypothesis behind the lasso distributions: on the left, the prior distribution followed by the coefficients, assuming the lasso hypothesis (Laplace distribution). On the right, a comparison of a Laplace and a Gaussian distribution.

The two most notable differences between Gaussian and Laplace distributions are:

1. A Laplace distribution has a probability spike at zero. This encourages cumulation of coefficients to be exactly at zero - hence permitting variable selection.
2. It is more likely to observe large values from a Laplace distribution than a Normal one, as Laplace has a heavier tail. As a consequence, large coefficients tend to be shrunk less in lasso penalty than in the ridge.

These two distributional differences have a direct effect on the lasso and the ridge estimates. Figure 2.5 compares the lasso and ridge coefficients on the multivariate VehBrand. As expected, lasso sets to zero some parameters and applies lower shrinkage to extreme parameters. We refer to Casotto, Banterle, and Beraud-Sudreau (2020) for a more in-depth comparison between the lasso and the ridge.

The Laplacian prior seems to incorporate the following statistical hypothesis: if a coefficient is not significant, it is set to zero; if it is above some significance threshold value, then the resulting estimate is still shrunk toward the complement of credibility zero.

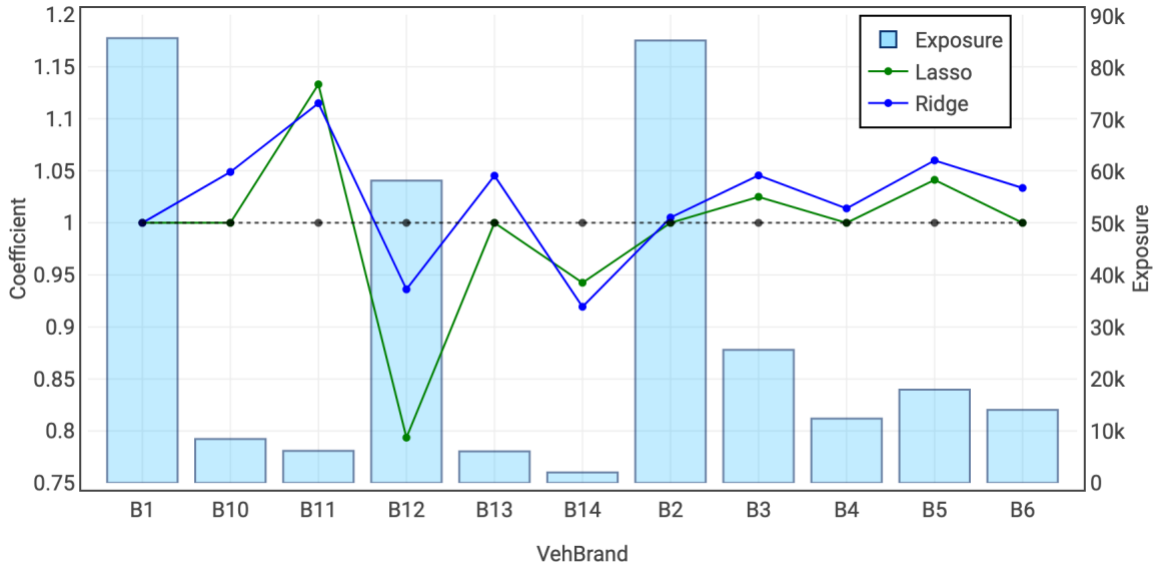



Figure 2.5: Comparison of ridge and lasso fits for the multivariate variable VehBrand, with same GLM formula as in Chapter 1.

## Comparison between lasso selection and p-values

From a practical perspective, the way that lasso penalty sets some coefficient to zero implicitly considers variable significance. We can illustrate such a statement by comparing a simple GLM fit with a lasso model fit using the same multivariate model analyzed in Section 1.3.2. During the multivariate modeling, a p-value analysis highlighted that some of the VehBrand parameters were not significant or borderline significant such as B14.

Figure 2.6 compares the **multivariate** fit of the lasso with the unpenalized GLM model for the VehBrand variable , showing that the lasso sets exactly to the base level some of the parameters.

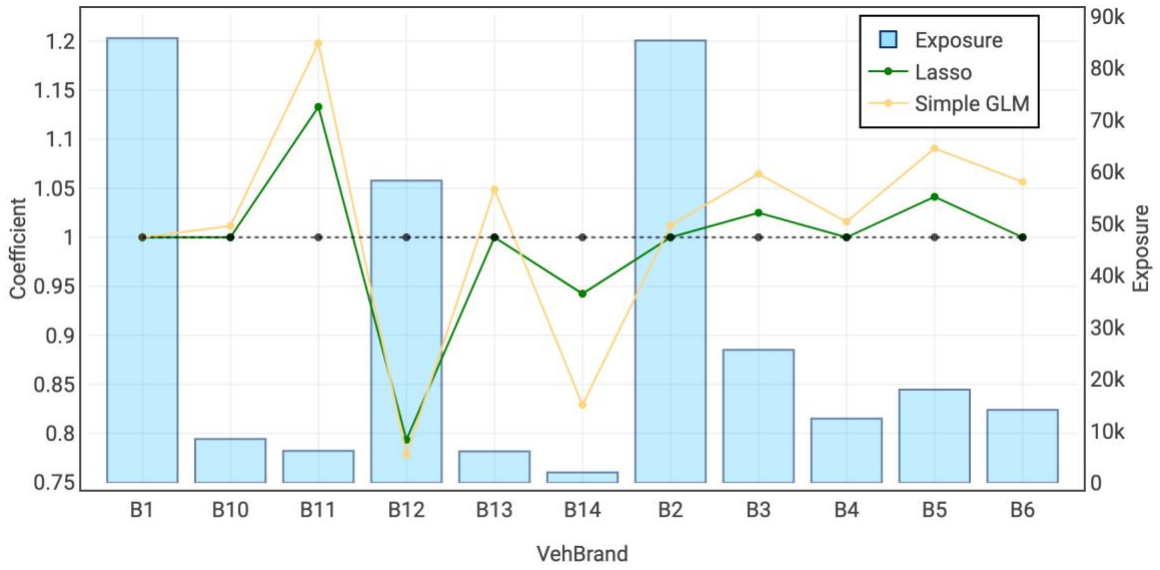



Figure 2.6: Comparison of lasso and GLM fit for the multivariate variable VehBrand, with same GLM formula as in Section 1.3. The lasso automatically disregards the parameters [B10,B13,B2,B4,B6], which are also not significant for a p-value analysis. The selected coefficients are shrunk: the lower the exposure, the larger the shrinkage. This is comparable to the effect of a credibility procedure whose complement of credibility is given by the grand average of the effect.

The table below displays both the p-value analysis and both GLM and lasso coefficients. In this example, the parameters that are equal to zero coincide with all those values whose p-value is below the 5% threshold. In a sense, the factor selection coincides with one modeler discarding all factors not deemed statistically significant. For the parameter B14, whose p-value was close to 5%, the lasso model is able to provide a credibility weighted estimate, which keeps the same directionality as the trend observed, hence being more appropriate than the original GLM estimate.

| VehBrand | GLM coefficient | Lasso coefficient | p-Value |
|---|---|---|---|
| B1 | 1.00000 | 1.00000 | nan |
| B10 | 1.01167 | 1.00000 | 0.78277 |
| B11 | 1.19804 | 1.13311 | 0.00005 |

| VehBrand | GLM coefficient | Lasso coefficient | p-Value |
|---|---|---|---|
| B12 | 0.77841 | 0.79358 | 0.00000 |
| B13 | 1.04876 | 1.00000 | 0.31601 |
| B14 | 0.82906 | 0.94246 | 0.04617 |
| B2 | 1.01219 | 1.00000 | 0.50507 |
| B3 | 1.06474 | 1.02499 | 0.01297 |
| B4 | 1.01589 | 1.00000 | 0.65058 |
| B5 | 1.09073 | 1.04132 | 0.00289 |
| B6 | 1.05653 | 1.00000 | 0.09113 |

This example illustrates the possibility offered by the lasso hypothesis to simultaneously select the non-significant parameters (like a p-value levels selection would do) and apply credibility to the significant yet weak effects captured in the models (like credibility would do).

# 3. Derivative lasso

## 3.1 Shortcomings of applying lasso to ordinal variables

The previous chapter highlighted how penalized regression, in particular Lasso regression, is a valid methodology to **blend credibility into a GLM**. Lasso regression achieves this by automatically eliminating non-significant parameters, resulting in more parsimonious models compared to simple, non-penalized GLM models. Additionally, Lasso's credibility assumptions are statistically sound due to the connection between penalized regression and Bayesian models, such as Buhlmann's.

There are strong benefits to use lasso to model **categorical variables**: in such a setting lasso's ability to select or shrink individual coefficients $\beta$ bears a direct actuarial interpretation. In categorical variables each individual level is assigned to an individual coefficient: when lasso sets to zero a coefficient, it means that the corresponding categorical level is not credible enough.

When modeling an **ordinal variable** or a continuous numeric variable, the actuarial meaning of the coefficients $\beta$ is less intuitive. When a variable is modeled linearly (such as variable Area), the value of the coefficient $\beta$ corresponds to the slope of the line used to describe the model. As shown in Figure 3.1, direct application of lasso penalty to the coefficient of the slope will result in a reduction of the slope value. When lasso sets to zero the coefficient, the linear trend will be flat hence resulting on the removal of the variable from the model.

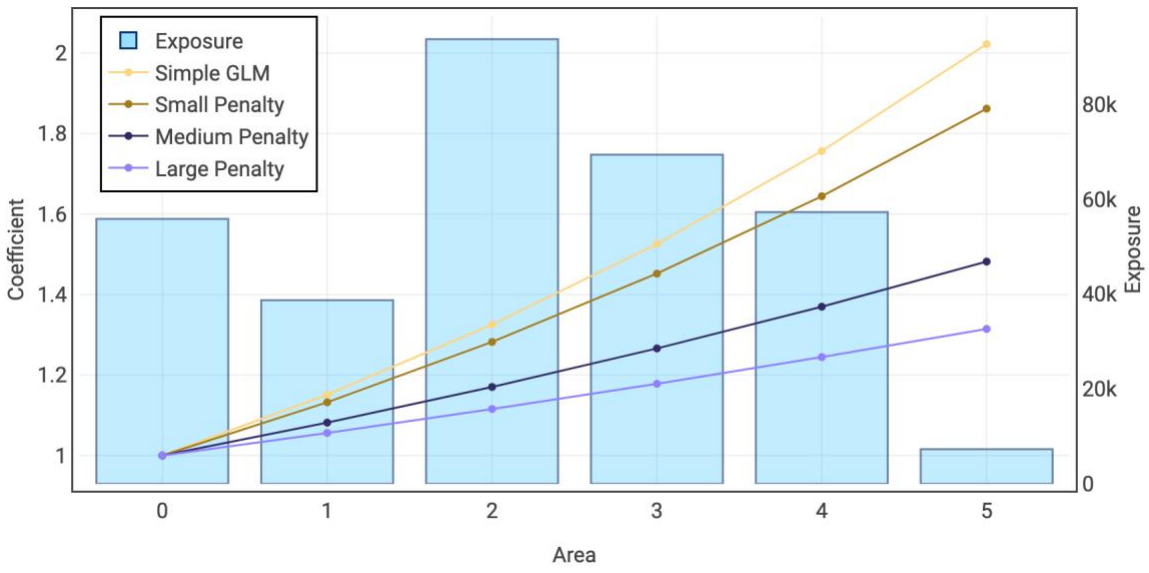



Figure 3.1: Lasso fit for various penalty values $\lambda$ on the univariate Area model ClaimNb ~ Area.

The coefficient in this example represents an increasing trend and is quite easy to understand. Characteristics that have more complex signal will require the modeler to engineer new

features to adequately capture this pattern. To visualize the predicted signal for this characteristic, a modeler must combine all of the engineered variables into a single curve. The more complex the feature engineering, the less intuitive the actuarial interpretation of each individual variable will be. Therefore, the effect of the shrinkage of each variable also becomes less intuitive as feature engineering becomes more complex. Consider the example of the polynomial representation of Driver Age:

```
ClaimNb ~ DrivAge + I(DrivAge**2) + I(DrivAge**3) + I(DrivAge**4) +
log(DrivAge)
```

If we assume that the lasso will remove the parameter $\beta$ associated to I(DrivAge**3), it is not immediately intuitive on how this will affect how the Driver Age overall trend. For such complex representations, lasso shrinkage and selection could still improve the models' performance, but the actuarial interpretation will be less explicit.

This lack of explicitness of lasso for ordinal variables, together with the requirements to still define the manual feature engineering (polynomial, hinges and others), shows that lasso's framework **alone** is not able to automatically detect and model nonlinearities for ordinal variables.

Fortunately, an intuitive change of the penalty structure allows to overcome lasso limitations. This chapter will show how moving from lasso to **derivative** lasso incorporates nonlinearities directly and provides a clear connection between the effect modeled and the structure of the penalty.

## 3.2 From lasso to derivative Lasso: An intuitive introduction

An alternative to modeling nonlinearities via feature transformations is to use one hot-encoding on an ordinal variable. Section 1.2.3 showed an example of how one hot-encoding can be used to model binning. We also showed that binning in a GLM has two main drawbacks: the continuity in the estimates is not guaranteed and variation within an interval is ignored.

These same drawbacks persist when applying lasso penalty to ordinal variables, as illustrated in Figure 3.2.

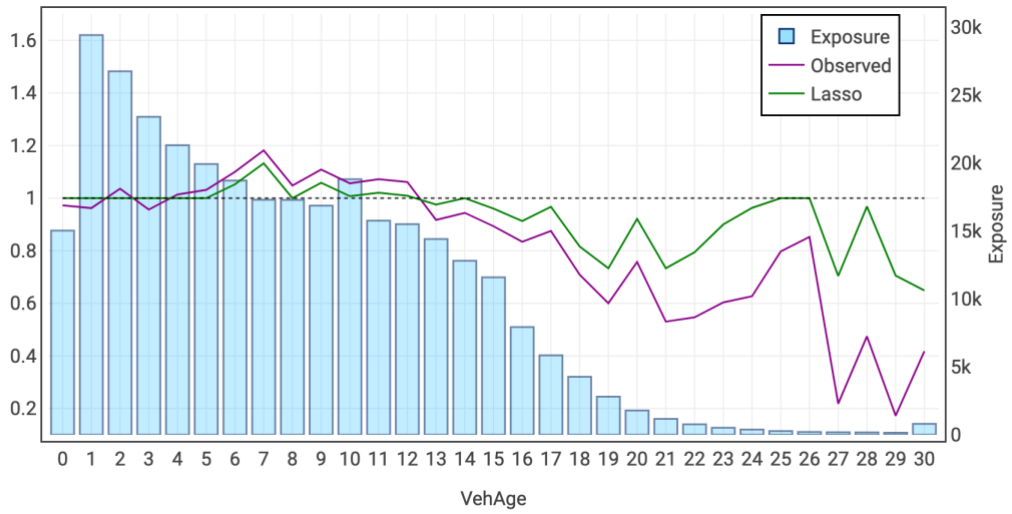



Figure 3.2: Lasso GLM result for the univariate VehAge variable, with formula ClaimNb ~ C(VehAge).

In the vehicle age case, the lasso coefficients do not fit very well to the underlying signal. This is because the underlying signal has some order to it, but the modeled coefficients are each independently shrunk towards 0. The “a priori” assumption that coefficients should be zero in the absence of signal results in a poor fit.

Figure 3.3 presents a visual representation of the lasso penalty’s influence. The lasso penalty acts like a “spring” connecting prior assumptions (black line) and coefficients (blue boxes). Each vehicle age is treated independently, and the model does not impose linearity constraints or penalties for reversals.

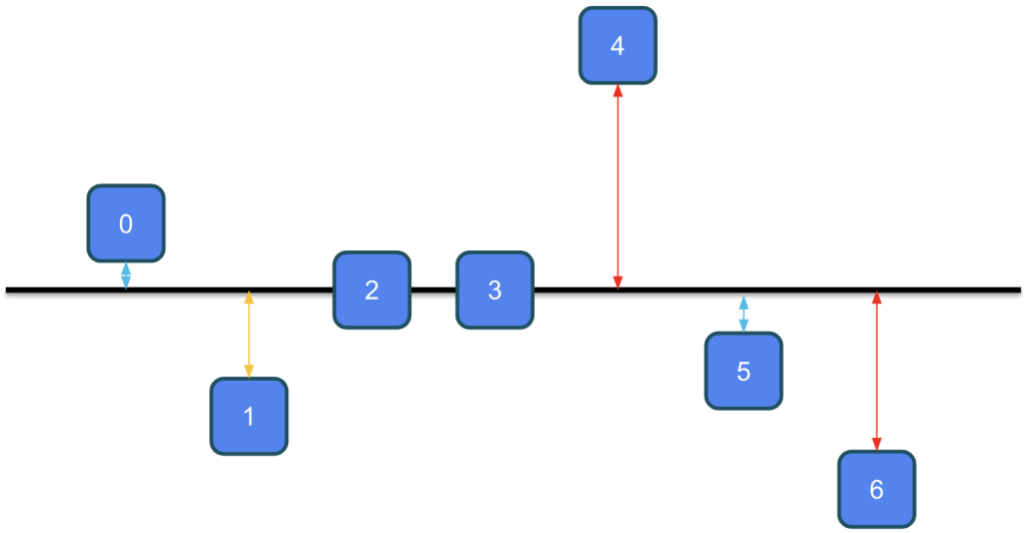



Figure 3.3: Schematic representation of the action of the lasso penalty, imposing a direct relationship between the coefficients (blue boxes) and priors (black line).

Intuitively, achieving continuity in estimates and model variation within intervals can be accomplished by directly connecting neighboring coefficients, rather than linking each of them separately to a global prior, as represented in Figure 3.4.

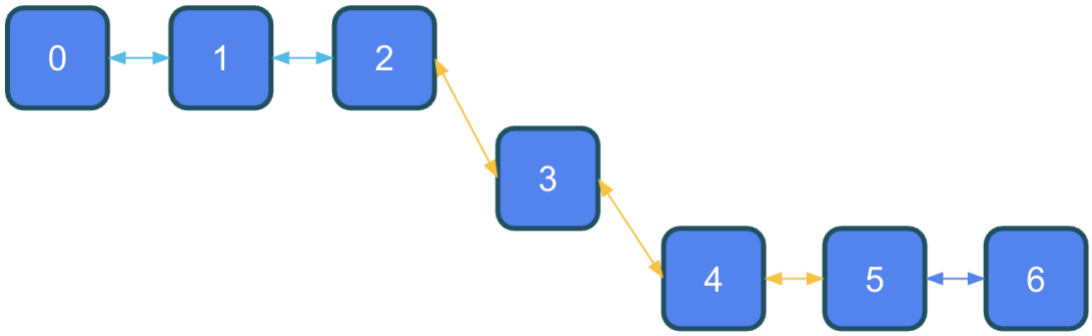



Figure 3.4: Schematic representation of desired relationship between coefficients for ordinal variables.

It is possible to achieve such a structure by choosing an appropriate prior hypothesis for the penalized regression. Once such a prior hypothesis is defined, we will show how the resulting penalty, the **derivative lasso**, tackles the challenge of natively including nonlinearities in a GLM.

### Defining "a priori" grouping

Penalized regression allows a modeler to easily incorporate assumptions in the model given a well-defined prior. We now aim to motivate one specific "a priori" hypothesis that will fit with the intuition behind Figure 3.4.

Let's start by recalling how categorical variables are treated differently between GLMs and lasso. In **GLMs**, the choice of including or excluding coefficients can be informed by a p-value test. Specifically, **the hypothesis being tested is** $\beta == 0$. For **lasso**, the null hypothesis $\beta == 0$ is directly implemented through penalization by adding a **Laplacian prior distribution centered at 0**. This prior assumption has various beneficial properties including its ability to set noncredible coefficients directly to zero. The application of this prior assumption parallels the decision making process in p-value significance testing.

When dealing with ordinal variables, specifically with binning, the hypotheses being tested change. For example, in Section 1.3.3, a grouping between the bins $\beta_{Age=[0-6)}$ and $\beta_{Age=[6-12)}$ was motivated by the results of p-value testing. While the "nominal" hypothesis testing was $\beta_{Age=[6-12)} = 0$, since the base level was $\beta_{Age=[0-6)}$, the real hypothesis being tested was $\beta_{Age=[6-12)} = \beta_{Age=[0-6)}$. Implicitly, the modeler was not trying to discern whether each individual bin was different from the overall average, but instead they were testing whether the **variation** of the underlying trend was significant.

This implicit change helps motivate a change to the **a priori** assumption of penalized regression for ordinal variables. Now, instead of trying to pull coefficients towards 0, we aim to pull coefficients towards each other. The flexibility of the penalized regression framework allows it to explicitly express such an "ordinal" hypothesis. It can do so by changing the a priori assumption from testing a significance of the absolute value of the parameters (as the individual factors in Vehicle Brand) to a credibility of the variation between ordinal levels (as we expect for Vehicle Age). This structure will result in data-driven binning where consecutive ordinal levels are not credibly different.

From a Bayesian perspective, an underlying prior distribution that would promote data-driven binning should satisfy two main properties:

1. Significance: if two consecutive coefficients are not credibly different, then they should be equal.
2. Stability: if two consecutive coefficients are credibly different, then they should be more likely close than far apart, so as to avoid unexpected wiggles of the trend.

This is equivalent to requiring that the **variation** between consecutive coefficients should be null if the variation is not credible and shrunk otherwise. With similar arguments as in Chapter 2, we can support that a Laplacian prior on the **differences** $\beta_i - \beta_{i-1}$ is appropriate. That is:

$$p_{prior} \sim e^{-\lambda|\beta_i - \beta_{i-1}|}$$

The associated penalty to such a prior is what we now define as the **derivative lasso**:

$$\beta = \underset{\beta}{\text{argmin}} \text{NLL}(y, X, \beta) - \sum_{i=1}^{30} \lambda \left|\beta_{VAge,i} - \beta_{VAge,i+1}\right|$$

In the remainder of this paper, we refer to the right penalty term as the **derivative lasso**, as it applies a lasso like penalty to the first order derivative of the categorical encoding of the VehAge variable.

## 3.3 The derivative lasso

Given one **ordinal** variable with $p$ levels, represented as $p$ different encoded coefficients $\beta = (\beta_1, \dots, \beta_p)$, we define the **derivative lasso** penalty to be

$$\text{Derivative Lasso}(\beta) = \sum_{j=1}^{p-1} \left|\beta_{j+1} - \beta_j\right|$$

Derivative lasso can be combined with other penalties in a modular way. For example, given one ordinal variable and one categorical variable, it is possible to combine lasso and derivative lasso via the penalized regression formula:

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \mathrm{NLL}(y, X, \beta) + \lambda\left(\mathrm{Lasso}(\beta_{\mathrm{Cat}}) + \mathrm{DerivativeLasso}(\beta_{\mathrm{Ord}})\right)$$

with $\beta_{\mathrm{Cat}}$ being the coefficients associated to the categorical variable, $\beta_{\mathrm{Ord}}$ for the ordinal variables.

The application of a lasso-like penalty to the consecutive differences of a categorically encoded variable has been already explored in the penalized regression literature. To the writer's knowledge, the first encounter of this penalty structure in the actuarial literature is in the paper of Semenovich (2013). Recently, this penalty has been popularized as a predictive modeling technique in the AGLM paper of Fujita et al. (2020). The AGLM paper presents several additional ways to extend the lasso penalty by means of feature transformations. Among these, we identify one of these feature transformations to be equivalent to the derivative lasso penalty. In a sense, the derivative lasso can be seen as an example of AGLM. Solvers for the derivative lasso penalized regression are also available on commercial software solutions such as Akur8.

### Role of the penalty parameter in derivative lasso

We will now illustrate how the derivative lasso penalty influences the coefficients of an ordinal variable. For this example, we encode an individual coefficient to each level of the Vehicle Age variable so that $\beta_{\mathrm{VehAge}} = (\beta_0, \ldots, \beta_{30})$. Figure 3.5 shows the solutions to the penalized regression with derivative lasso penalty for various values of the penalty $\lambda$.

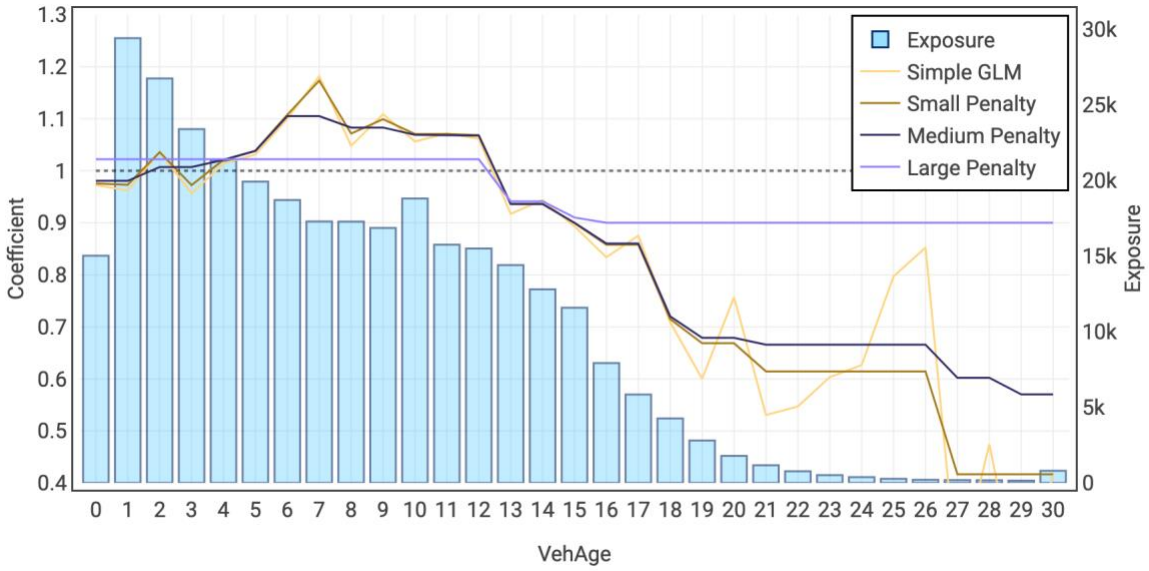



Figure 3.5: Derivative lasso fit for various penalty values $\lambda$ on the univariate Vehicle Age model. Simple GLM represents the unpenalized GLM coefficients to a categorical treatment of the Vehicle Age variable. The constant black dashed line at 1.0 illustrates an equivalent to the prior assumption of the derivative lasso.

The penalty parameter in the derivative lasso acts as a "smoothing" parameter: the higher the penalty, the more grouped or "smooth" the underlying trend appears. More precisely, different values of $\lambda$ correspond to:

- **Small** $\lambda$ (small penalty)
  - Coefficients are **close** to the simple, unpenalized GLM (actually equal to the GLM when $\lambda = 0$).
  - Levels with very low exposures tend to be grouped, but we may still observe a large amount of trend reversals due to noise on observed data (which are present in the unpenalized, simple GLM estimate).
  - A small penalty results in a volatile model with a tendency to overfit to the modeling data.
- **Medium** $\lambda$ (medium penalty)
  - Coefficients follow a proper tradeoff between the observed data (simple GLM) and responsiveness.
  - The trend described by the coefficient appears to be sound. The only trend reversal, from levels 5 to 14, seems to be justified by the data.
- **Large** $\lambda$ (large penalty)
  - Coefficients are close to the complement of credibility.
  - The coefficients are coarsely grouped, thus reducing the segmentation power of the variable.
  - A large penalty results in an overly robust model with a tendency to underfit model the modeling data.

### Comparison between derivative lasso and GLMs

Figure 3.10 compares the derivative lasso solution with the GLM coefficients showed in Section 1.2.3.

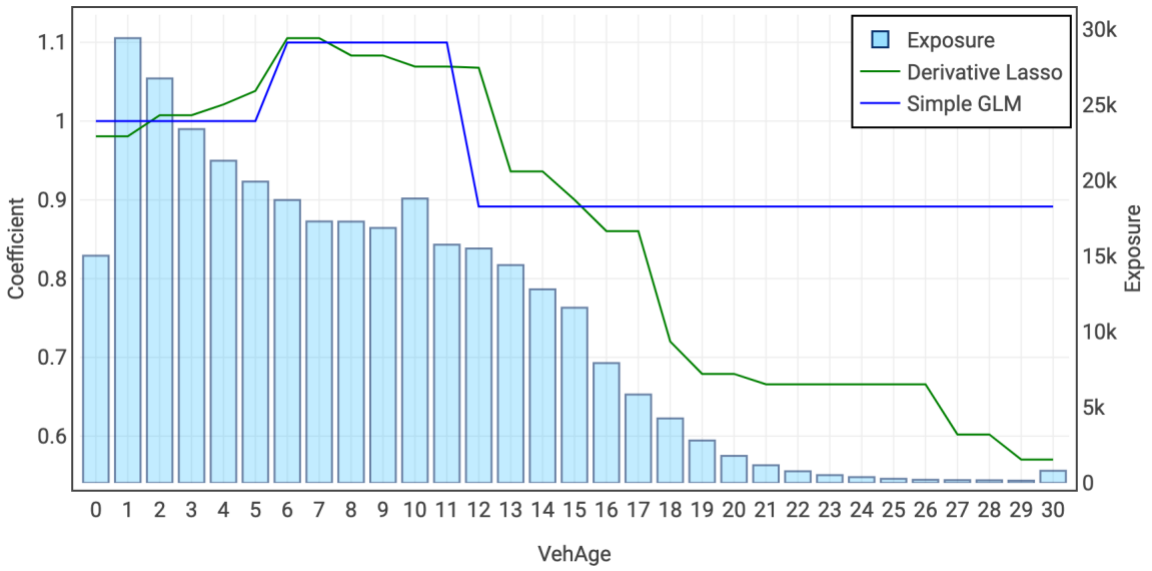



Figure 3.6: Comparison of penalized GLM coefficients (green) and the simple, non-penalized GLM coefficient described in Section 1.3.3 for the Vehicle Age variable.

We highlight three main properties of the derivative lasso solution:

1. **Adaptive grouping**: In the GLM example in Chapter 1, the modeler had to specify the grouping of the variable **before** the GLM fit. In the derivative lasso, the user did not specify the resulting groupings, but assigned one individual parameter for each vehicle age, as done with categorical variables. The resulting grouping is enforced by the derivative lasso penalty. We observe that the overall grouping is a result of the tradeoff between deviations on the observed values and number of underlying exposures.

2. **Trend following**: When the underlying observed trend is monotonically decreasing, the derivative lasso estimates will follow such a monotonic trend. Similarly, derivative lasso estimates will be stable when the observed trend is not clearly monotonic.

3. **Credibility**: The segment from 27 to 30 years old has limited data. The underlying trend is still monotonic, and the resulting binning is relatively granular. We stress that the resulting estimate for the bin is shrunk toward the overall mean of the observations, as one can determine by observing the shift from the coefficients and the observed values. Derivative lasso, similar to ridge and lasso, return estimates that are credible and therefore tend to be shrunk toward the grand average.

### Advantages of level binning with derivative lasso

These three properties shows how the derivative lasso solves simultaneously both of the major drawbacks of binning procedures illustrated in Section 1.2.3 :

- **Continuity in the estimates is not guaranteed**: Goldburd et al. (2016) states that “there is no way to force the GLM to have the [grouped] estimates behave in any continuous fashion, and each [grouped] estimate is derived independently of the others”.

Conversely, the prior induced by derivative lasso forces a dependency between consecutive levels, so that each estimate is not derived independently, but instead depends on their neighboring estimates. Furthermore, by modeling only the statistically significant deviations, the penalty prevents noise from creating the "wiggling" behavior that is observed on a GLM's estimate of bins with limited exposures.

- **Variation within intervals is ignored**: Goldburd et al. (2016) states that "Since each bin is assigned a single estimate, the model ignores any variation in severity that may exist within the bins. […] Of course, we could refine the interval boundaries to split that bin into two or more smaller ones. Doing so, however, would thin out the data, reducing the credibility of the estimates, thereby making them more susceptible to noise.". The limitations of the simple, unpenalized GLM require the modeler to define the bins of the ordinal variable by a top-down approach, where coarse bins are investigated first, and then refined into smaller ones until the estimates are not credible anymore. The derivative lasso allows for a more efficient bottom-up approach: each individual value is modeled as an individual bin, letting the induced prior to adaptively merge these bins. The benefit of this approach is also motivated by the theory, as the derivative lasso achieves the optimal rate of convergence over a broad class of underlying functions [Tibshirani (2014)].

## 3.4 Application of derivative lasso to multivariate analysis

Chapter 2 demonstrated the suitability of the lasso penalty for categorical variables like VehBrand. In this section, we illustrate how the derivative lasso penalty can provide a reliable statistical assumption for ordinal variables such as VehAge and DrivAge.

We propose that the combination of these two penalties offers a powerful approach for multivariate modeling. **Ordinal** variables will use the **derivative lasso** penalization, while **categorical** variables will continue to use the **lasso** penalty term. To support this claim, we compare the output of a multivariate GLM model presented in Section 1.3 with that of a penalized model that employs the lasso and derivative lasso penalties, depending on the nature of the variables.

It is important to emphasize that the **penalized model does not require any prior variable transformations** to be applied to the data used in the modeling process. This represents a major improvement over the simple, unpenalized GLM procedure, which often involves a time-consuming and iterative process of transforming individual variables through techniques such as binning, variable transformations, and parameter selection in order to develop an appropriate model.

Instead, the penalized regression approach provides a more efficient and robust alternative by incorporating the intended structure of the model as a statistical assumption within the fitting procedure. This not only saves significant time, but also increases the robustness of the underlying trend being described by the model.

To illustrate this approach, we introduce a custom "formula representation" for the penalized multivariate model. This formula is for illustrative purposes only.

```
ClaimNb ~ C(Region) + C(VehBrand) + C(VehGas) +
          D(Area) + D(BonusMalus) + D(VehPower) +
          D(VehAge) + D(DrivAge)
```

To represent the one-hot encoding procedure, we use the C (contrast) notation, as in the standard formula notation provided in the 'glm' R package. The relative encoded parameters $\beta$ are penalized with a lasso penalty of $|\beta|$.

For ordinal variables in the freMtpl dataset, we use the custom D notation (D as in derivative) to indicate that these variables are represented via the categorical encoding procedure. However, the parameters $\beta$ are penalized with a derivative lasso penalty of $|\beta_i - \beta_{i-1}|$.

The penalty parameter $\lambda$ used to fit the penalized model is chosen according to the procedures and considerations defined in Chapter 4.

### The Vehicle Brand variable

We begin our analysis by considering the categorical variable VehBrand, which is included in the model with a penalty similar to the lasso. Figure 3.7 shows the quality of the model fit, while Figure 3.8 compares the estimates produced by the penalized model with those produced by the GLM model described in Chapter 1.

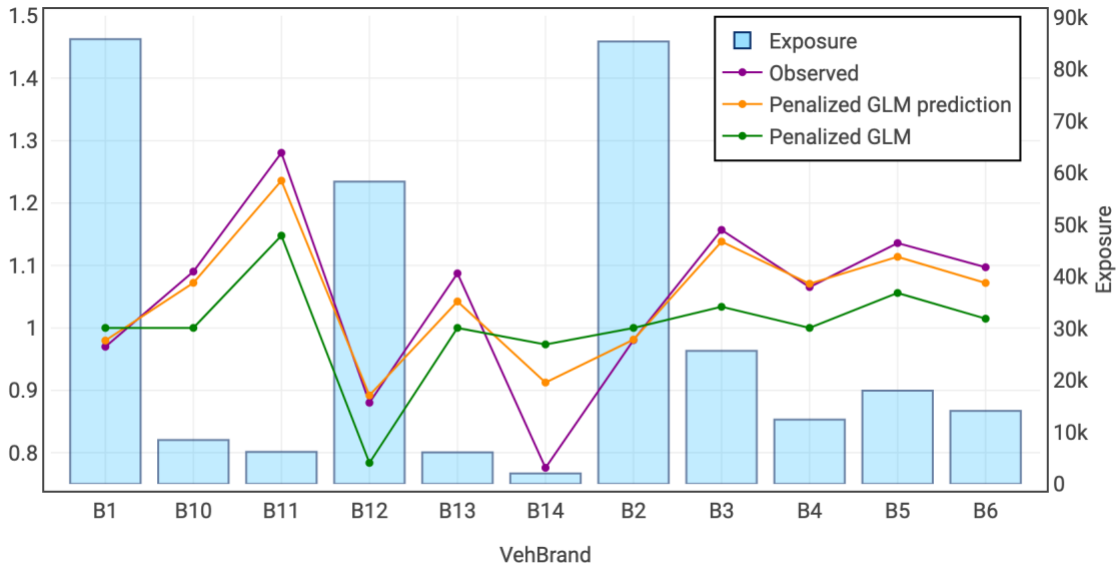



Figure 3.7: Plot of observed claims (purple), multivariate penalized GLM predictions (orange) and coefficients (green) for the Vehicle Brand variable.

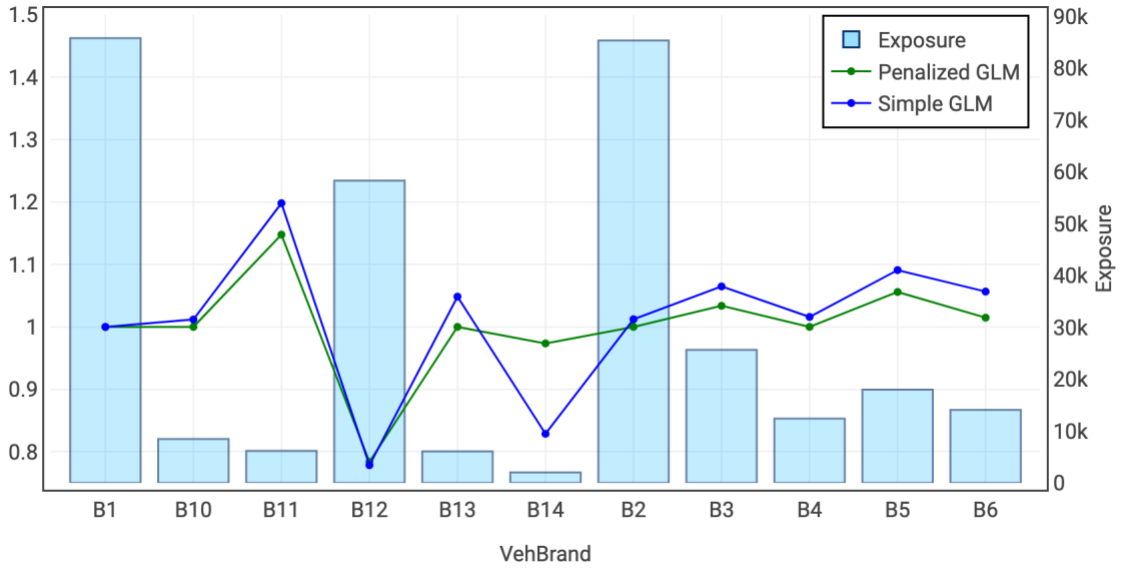



Figure 3.8: Comparison of penalized GLM coefficients (green) and the simple, unpenalized GLM coefficient described in Section 1.3.2 for the Vehicle Brand variable.

As in the lasso example from Section 2.4, the brands with indices 2, 4, 6, 10, and 13 are deemed noncredible by the penalty. The penalized model yields estimates that agree with a GLM p-value analysis, despite the differences between the two models. For all selected parameters, the resulting estimates are shrunk as in a credibility method.

### The Vehicle Age variable

We now proceed to analyze the ordinal variable Vehicle Age, whose penalty is the derivative lasso.

Figure 3.9 represents the quality of the fit of the model and Figure 3.10 compares the penalized estimates with the GLM model described in Section 1.3.

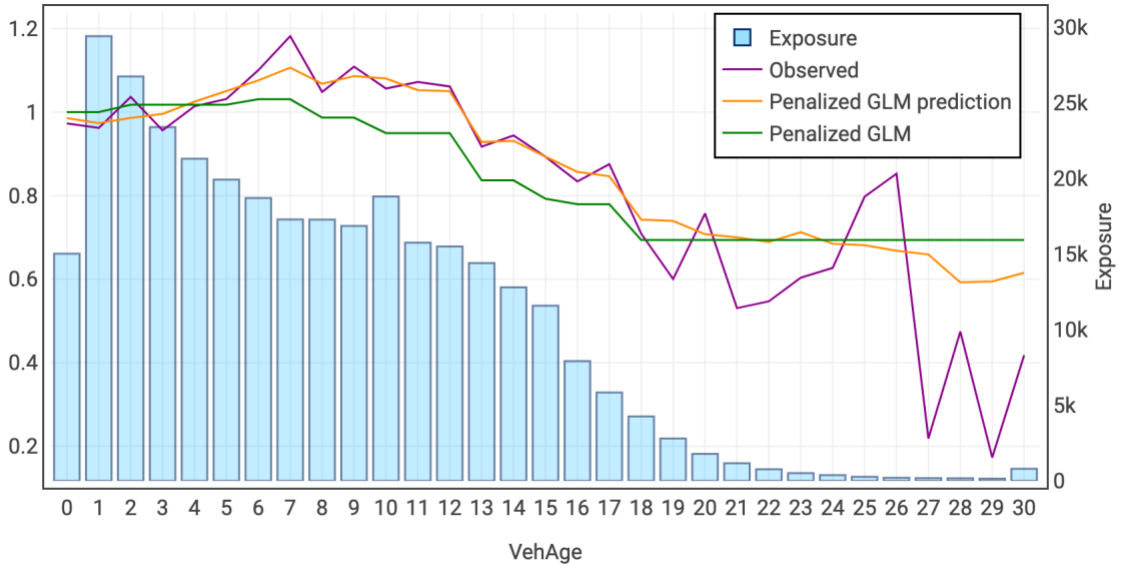



Figure 3.9: Plot of observed claims (purple), multivariate penalized GLM predictions (orange) and coefficients (green) for the Vehicle Age variable.

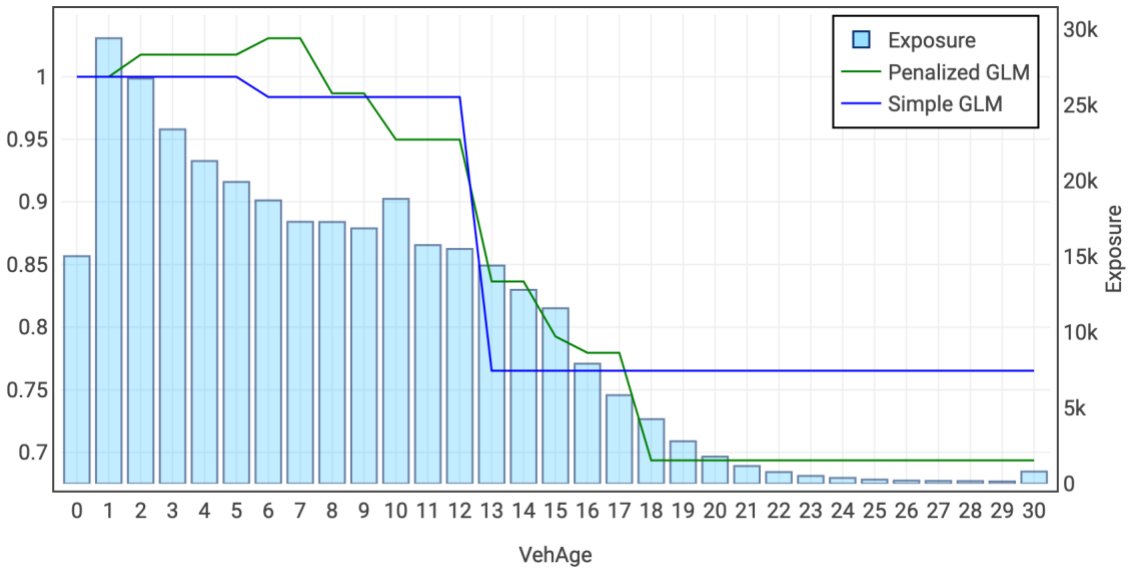



Figure 3.10: Comparison of penalized GLM coefficients (green) and the simple, non-penalized GLM coefficient described in Section 1.3.3 for the Vehicle Age variable.

Figure 3.9 shows that the derivative lasso penalty improves the fit of the Vehicle Age model. By using a data-adaptive binning method, the derivative lasso is able to interpolate the penalized solution and produce a more refined trend than could be achieved through iterative refinement of residuals. This is especially noticeable for ages between 6 and 18, where the derivative lasso is able to determine credible groups, whose estimated coefficients follow a decreasing trend.

Determining such a fine grouping using a simple, unpenalized GLM would have required a large number of cumbersome iterations due to the exponential number of possible combinations of grouping.

### The Driver Age variable

In Section 1.2.4, Driver Age was modeled as a combination of polynomials and a logarithmic function to create a continuous function. The derivative lasso penalty produces a model that captures the inherent nonlinearity of the fit with good quality.

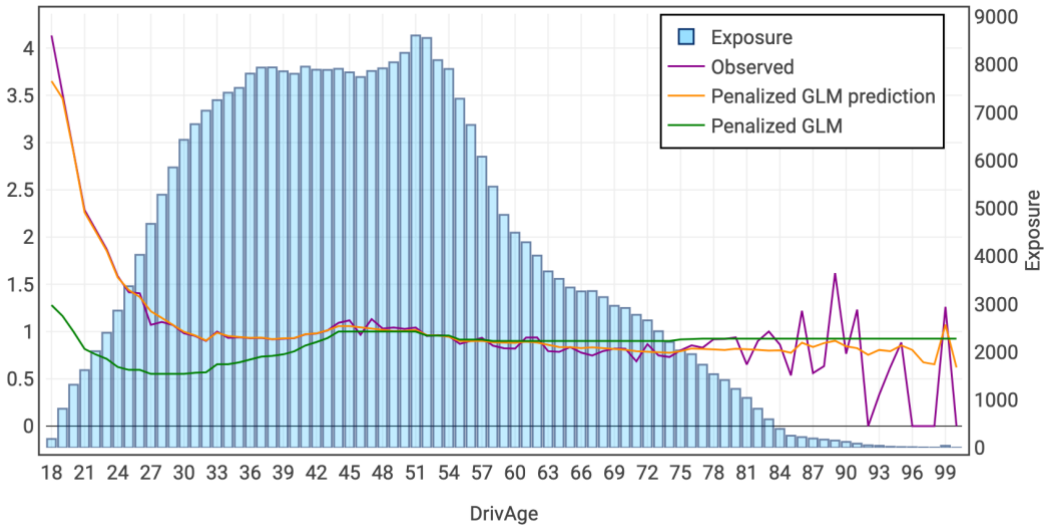



Figure 3.11: Plot of observed claims (purple), multivariate penalized GLM predictions (orange) and coefficients (green) for the Driver Age variable.

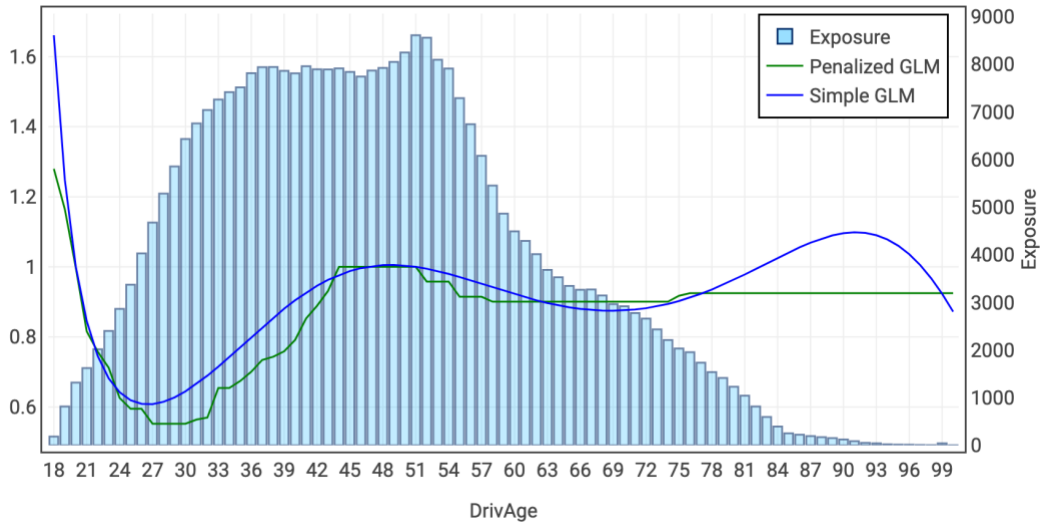



Figure 3.12: Comparison of penalized GLM coefficients (green) and the simple, non-penalized GLM coefficient described in Section 1.3.4 for the Driver Age variable.

Derivative lasso results on much more stable estimates on the tails of the curve than the simple, unpenalized glm model. We see a significant difference is observed in the trend for the older population above 78. The GLM model produces a trend that increases until ages 93 and then abruptly decreases, which is often observed when using polynomials. This “whip” effect is a consequence of trying to force expected behaviors on other segments of the population, particularly the sharp decrease in risk around age 27. This behavior is a result of the parameterization creating a good fit where there are many exposures at the expense of a poor fit where there are fewer exposures. This tradeoff maximizes the GLM penalty optimization of likelihood but creates an unintuitive factor curve for older driver ages.

In contrast, the penalized GLM estimates are agnostic to this second-order effect and rely on a sounder statistical assumption. If there is insufficient evidence to support a change in the trend, then the trend should remain constant. This approach results in a more accurate and fairer estimate of the risk for the older generation, which is approximately constant.

The penalized GLM identifies a steeper decrease in the risk for ages between 25 and 45 than the GLM estimates, highlighting the additional flexibility of the nonlinear trends modeled by the derivative lasso.

This additional flexibility is expected. The derivative lasso changes the modeling from a “parametric” description (for example, a combination of polynomials) to a “non-parametric” approach. A priori, we model all possible functions, and we choose the one that is most appropriate to our prior assumptions.

### The Area and Density variables

An unexpected result of the multivariate penalized GLM variable is the removal of the variable Area, as depicted in Figure 3.13.

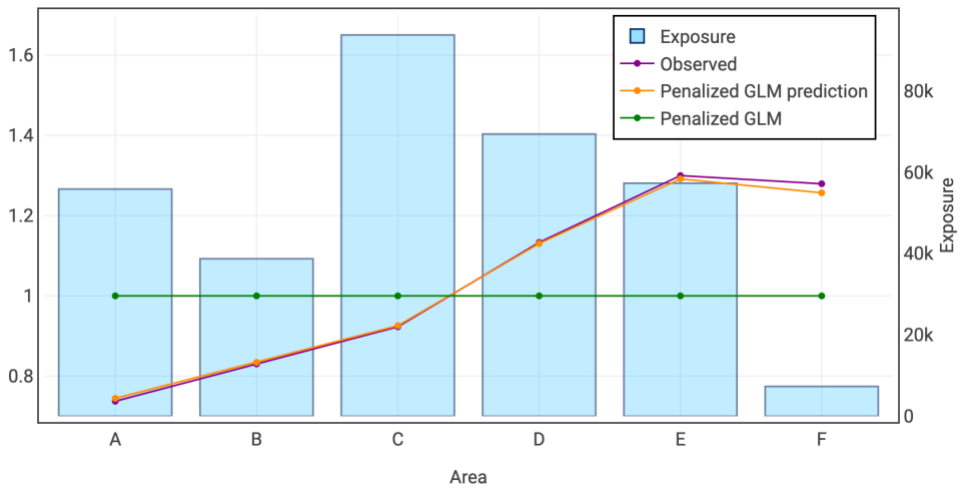



Figure 3.13: Plot of observed claims (purple) and multivariate penalized GLM predictions (orange) for Area variable. Since the derivative lasso removed the variable from the model, the coefficients (green) are constantly equal to 1.0.

We observe that, even if the Area variable is not modeled, both the observed and the prediction are close, meaning that the signal of the variable Area is captured by other variables.

A closer look at the data shows that the high correlation between the variable Area and the Density variable, is in fact, a dependence. The variable Area is a grouping of the Density variable with cut points at 50, 100, 500, 2K and 10K.

The exclusion of the Area variable may be explained by the data-driven binning of the Density variable done by the derivative lasso. Figure 3.14 illustrates the Density variable, grouped by the bins defined by the derivative lasso fit. The resulting data-driven segmentation is more refined than the one used in the Area, which results in a removal of the now insignificant Area variable.

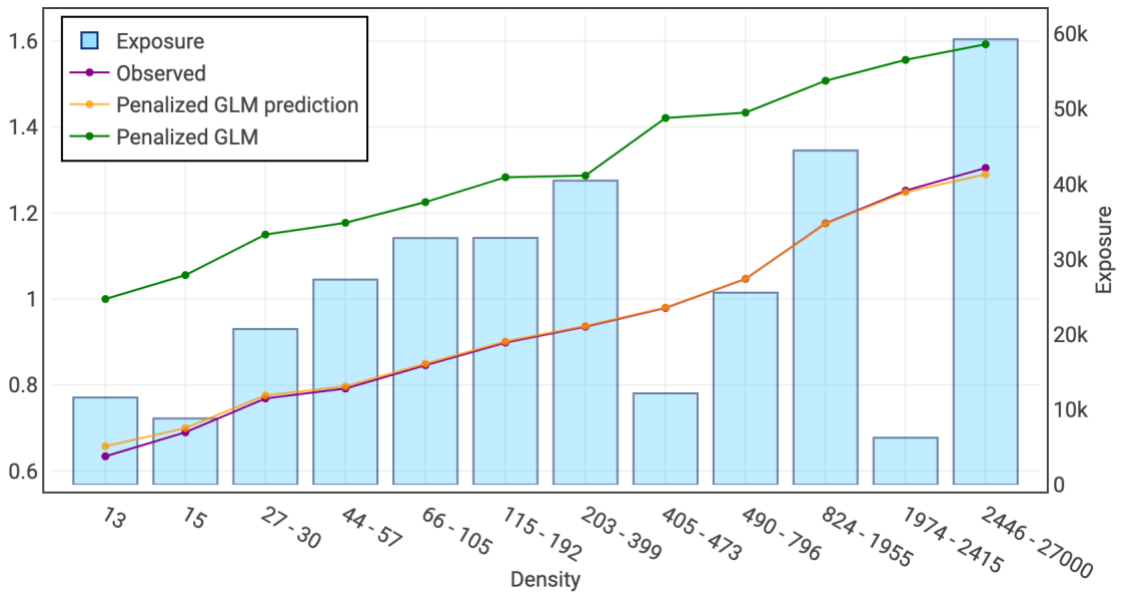



Figure 3.14: Plot of observed claims (purple), multivariate penalized GLM predictions (orange) and coefficients (green) for the Vehicle Age variable. The bins represent the natural grouping induced by the derivative lasso penalty.

# 4. Model selection and validation

The previous chapters provided a holistic perspective on penalized GLMs, with a particular focus on lasso and derivative lasso penalties. The reader should now be familiar with the connections between penalized and unpenalized GLMs and credibility, and how the choice of the penalty parameter influences the robustness of the resulting models. The focus of this chapter is to provide the reader with a process to leverage penalized GLMs to build predictive models with any insurance dataset.

We will illustrate how such a process is divided into two main phases. The first phase, which we will refer to as the **data-driven phase**, aims to define a reasonable penalty parameter interval that is suitable for both the data and the modeler's task.

After the data-driven step, the modeler will move onto an **expert-driven phase**, where the modeler will validate the interval identified in the previous step and, within this interval, select the appropriate level of penalty based on the models' structure. This step will share many similarities with the current GLM process of validation, which will be highlighted in the next section. In particular, it will be possible to apply actuarial judgment to the model thanks to the table-like output of penalized GLMs.

This two-step procedure integrates with and expands the model construction guidelines for a simple GLM. Even if we assume the reader is familiar with these procedures, we will quickly summarize the evaluation best practices in the following section for the sake of clarity.

## 4.1 Model selection and validation for unpenalized GLMs

Building GLMs requires a modeler to make many feature engineering modeling decisions, and for each choice there is a huge number of other alternatives. Examples include choosing the degree of polynomial, as in Section 1.2.4, or removal of a factor, as in Section 1.3.2 . There are several ways to support such decisions in standard GLM modeling.

Chapter 1 illustrated how visualization of the variables and significance tests can support a modeler's decision-making. Another fundamental dimension to consider is the ability of the model to generalize correctly with unseen data. Model fit is generally evaluated using metrics such as Gini and deviance (or pseudo R2[6]). Calculating these metrics in the same dataset

[6] Pseudo R2 is a transformation of the deviance to a percentage.

$$\text{PseudoR2} = 1 - \frac{\text{Deviance(model)} - \text{Deviance(average)}}{\text{Deviance(saturated model)}}$$

When Pseudo R2 is equal to 0, the model performs equally to the average, while a value of 1 is equivalent to a model that perfectly replicates the data.

used to build the model will provide an overly optimistic assessment of the generalization power since the result will always favor the most complex model [Goldburd et al. (2016)].

It is often a requirement to split the dataset in two subsets: one will be used to fit the model (the **modeling dataset**), and the other will be used only to evaluate the performance of the model (the **holdout dataset**). It is extremely important to avoid using the holdout dataset during the creation of the model structure, because the performance of the model on that dataset will be used to measure the ability of the model to generalize on unseen data. Iterating (and thus using the holdout dataset to drive modeling choices) will implicitly integrate the observations of the holdout dataset into the model. The model's performance on new truly unseen data will then be different from the performance measured on the holdout dataset.

There are other strategies to verify that GLM model iterations will generalize well with good predictive power on unseen data without looking at the holdout dataset.

One approach consists of further splitting the modeling database into two parts, one to fit the model and the other to evaluate the predictive power of the model throughout the GLM iterations. For example, a modeler can decide between two feature engineering alternatives by fitting the model on the first portion of the data and to select the alternative leading to the best metric on the second portion, without using the holdout dataset. We will refer to these partitions of the modeling dataset as train dataset and test dataset respectively. Figure 4.1 visually describes the difference between the modeling-validation split and the train-test-validation split. The reader may find the choice of the partition names different from their experience. This can be explained by the lack of strict naming conventions for these partitions, as illustrated by the informal poll done by the NAIC at a 2022 CAS RPM presentation whose results are shown in Figure 4.1.

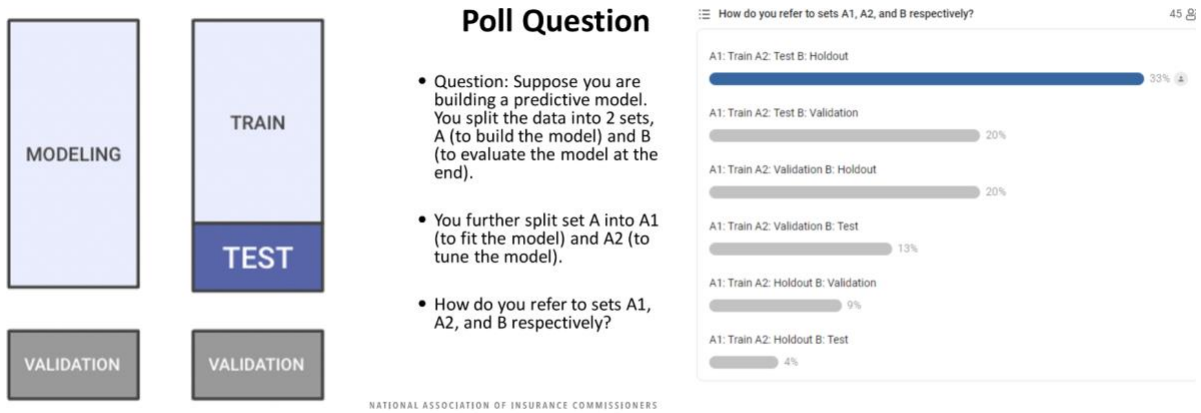



Figure 4.1: Panel (a) provides a visual description of both the modeling and validation split for final model evaluation and the train, test and validation split. Panel (b) represents an informal poll made at CAS RPM 2022 illustrating the lack of unique naming for the data splits. Figure taken from the NAIC's presentation "Increasing Speed to Market".

While this strategy provides, in theory, a good way to ensure the generalization power of the models created, it is very hard to strictly follow it in an operational environment where errors are identified, or feedback is given late in the process. If corrections are made after testing the models on the holdout datasets, these holdout datasets are "spoiled" for modeling purposes. It also requires a very strict discipline to ensure that the holdout datasets are used purely to assess the model performance and not to directly take the decision to accept or reject individual variables or levels. Such usage would prevent evaluation of a model's ability to generalize to unseen data.

Another way to compare models is to not use any testing set at all, but instead use statistical theory to approximate the generalization power of a model using training data alone. This approach does not use any test or validation datasets to assess the generalization power of the model, avoiding the problems described above. The Akaike Information Criterion (AIC) is one of the most popular of such metrics, whose formula is

$$\text{AIC}(\beta) = 2\text{NLL}(\beta) + \#\text{of degrees of freedom.}$$

Here, the NLL is the negative log-likelihood computed in the training data, and the number of degrees of freedom is the number of non-zero entries of the GLM coefficients $\beta$.

Given two different modeling alternatives, the modeler can compute the AIC for each model. Since the AIC formula "penalizes" the decrease of the NLL of the more complex model with the degrees of freedom (hence the complexity), the model with lower AIC should be preferred. The curious reader may notice that the AIC formula looks similar to a penalized GLM Equation 2.1, as they both consider the tradeoff between the goodness of fit (negative log likelihood) and the complexity of the model (in the AIC, the number of degrees of freedom). The connections between AIC and penalized GLMs, with lasso in particular, are deeper. For example, the penalized GLM version of the AIC is known as the "best subset selection" problem, which is proven to be numerically intractable. However, one can prove that the best "numerically tractable"[7] approximation to that problem is the lasso. The interested reader can refer to the work of Hastie, Tibshirani, and Tibshirani (2020) for a more in-depth discussion, and how lasso outperforms best subset selection in conditions similar to insurance data.

## 4.2 Model selection for penalized GLMs

Penalized GLMs, and in particular lasso and derivative lasso, offer the opportunity to significantly reduce the number of iterations when developing a GLM model via the choice of the penalty parameter value $\lambda$. For ease of exposition, we will refer to this parameter as the **smoothness** parameter.

[7] A problem is considered as numerically tractable if an optimization algorithm can provide its optimal solution in a reasonable time. This usually means that the problem is convex.

In the previous chapters (Chapter 2 for lasso and Chapter 3 for derivative lasso), we showed how the choice of the smoothness parameter value affects the properties of the created models:

- A **high smoothness value** results in **highly robust models**, which may, at times, fail to capture enough of the underlying signal.
- Conversely, a **low smoothness value** yields **very sensitive models**, which can be excessively volatile and prone to overfitting the data.

It is the modeler's responsibility to determine the appropriate smoothness parameter value in order to create a model that generalizes well to unseen data. Consequently, the selection of the smoothness parameter becomes a crucial aspect of the modeling process itself. It is important to note that there are no "out-of-the-box" formulas for estimating a single best smoothness parameter. For instance, adapting the formula for estimating the Buhlmann parameter $K$ to the penalized GLM framework is not feasible[8]. Moreover, the most suitable smoothness parameter can vary significantly from one dataset to another. Therefore, the practitioner should not focus on the absolute magnitude of the parameter but rather on the coefficients of the model itself, within a range of smoothness value guided by statistical analysis.

Last but not least, we will emphasize that the final choice of the smoothness parameter value is an exercise of **actuarial judgment**. Therefore, by the very definition of actuarial judgment, such a process cannot be entirely automated by a single formula or procedure. We will now describe a process for finding an appropriate smoothness level that takes into account both statistical and actuarial considerations. This procedure builds upon the GLM best practices mentioned earlier and is divided into two parts:

1. The **data-driven step** identifies a small interval of smoothness parameter values that yield high generalization power.
2. The **expert-driven step** focuses on the analysis of the coefficients of the penalized GLMs, incorporating actuarial and expert judgment to evaluate the coefficients of the models within this interval.

### The data-driven step: Cross-validation

In the data-driven step, the modeler begins with a broad range of smoothness parameter values and gradually refines this exploration to arrive at a smaller smoothness interval. These

[8] The formulas to estimate $K$ require estimating the variance within a class. Now, in a univariate setting as in Buhlmann credibility, such estimator is robust as the number of individual classes still has enough exposure. In a multivariate setting, the number of possible combinations of attributes grows exponentially, which irremediably degrades the quality of the estimator.

values will yield a collection of high-performing models, which will then undergo manual refinement by the modeler in what we refer to as the expert-driven phase.

We begin the procedure by considering a broad range of smoothness values. For example, we might use a set of log-spaced smoothness values that range from $\lambda = 0$ (where the penalty has no effect) to a high value where the penalty is so strong that $\beta = 0$ becomes the solution. The definition of what constitutes a "high value" is beyond the scope of this paper and is not a primary concern for the modeler. Most penalized GLM software provides out-of-the-box routines to identify such intervals.

Starting with this broad interval, the model (or software) will then calculate an approximation of the generalization power for each penalized GLM model with a specific $\lambda$ value. This can be accomplished through the "cross-validation" procedure illustrated in Figure 4.2. Cross-validation can be viewed as a generalization of both the train-validation-test split approach and the AIC procedure described in the simple GLM model selection in Section 4.1.

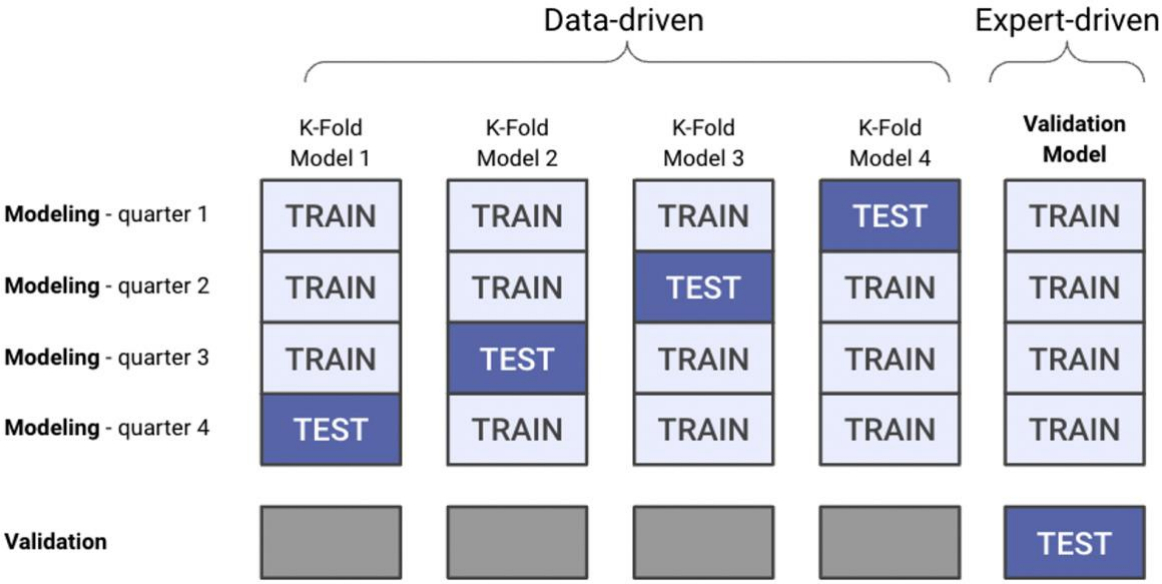



Figure 4.2: Visualization of the complete cross-validation procedure with $K = 4$ folds. The data-driven step will iteratively split the modeling dataset in 4 parts, and an equal number of models will be fitted on the relative partitions. The final model, which will be analyzed via actuarial judgment, is the one using all available modeling data, as highlighted by the expert-driven section of the figure.

Cross-validation provides a means of assessing a model's performance on unseen data through multiple train-test splits [Goldburd et al. (2016)]. Assuming the modeler has already divided the entire database into both a modeling and validation set, the general procedure for cross-validation consists of the following steps:

1. Split the modeling dataset into $K$ different, approximately equally sized groups referred to as folds.

2. For each unique fold:
    a. Treat that fold as a test dataset.
    b. Treat the remaining folds as a (sub) training dataset.
3. Fit a model on the training set and assess its performance (Gini or pseudo R2) on the test set.
4. Average the performance across the $K$ estimates to obtain a robust performance estimate.

The process of applying the cross-validation procedure while iterating through all the distinct values of the smoothness parameter is commonly known as a "grid search" in machine learning literature. By definition, the overall procedure is more computationally demanding than fitting a single unpenalized or penalized GLM. For instance, when testing 7 smoothness parameter values using a 4-fold cross-validation, it necessitates the computation of 28 different models. For each of these 28 models, performance metrics are assessed on the corresponding testing fold and then stored[9].

In standard machine learning applications, the final selected parameter value(s) is the one that produces the best cross-validation performance. This chosen parameter(s) is then used to fit the data on the full modeling dataset, resulting in the final model. However, when it comes to penalized GLMs in actuarial modeling, this procedure can and should be further enhanced.

Optimizing the model parameter value(s) based solely on cross-validation results is a necessity when dealing with the selection of multiple parameters, rather than just one. For instance, in the case of Gradient Boosting Machines (GBMs), creating a model involves finding the right combination of parameter values, such as the learning rate, the number of trees, and the tree depth. These parameters interact with each other, and manually adjusting a single parameter might not lead to a straightforward improvement in the model. This is the situation when one unavoidably needs to rely on cross-validation for parameter selection.

In the case of lasso and derivative lasso, the selection process is simplified as it involves choosing only the smoothness parameter. This single parameter is rooted in credibility, making it straightforward to understand the effects of increasing or decreasing its value. Thus, the modeler has one well-defined parameter to exercise actuarial judgment. Actuarial judgment can be applied to select the most actuarially appropriate model from within a range of statistically appropriate levels of smoothness.

[9] The computational cost of cross-validation can be significantly reduced via various strategies, such as parallelization and/or the usage of previous results as warm starts.

Figure 4.3 presents the cross-validation results for the multivariate application of derivative lasso discussed in Section 3.4. It showcases a series of smoothness parameter values that are equally spaced on a logarithmic scale. In the figure, the red line denotes the scores representing the average of the cross-validation scores, while the whiskers illustrate the deviation of the scores within each fold. The vertical blue line indicates the smoothness parameter that results in the best cross-validation metric. Notably, for the neighboring smoothness values, the model performances are quite similar. Considering the variations within the folds, the highlighted interval exhibits models with approximately equal performances.

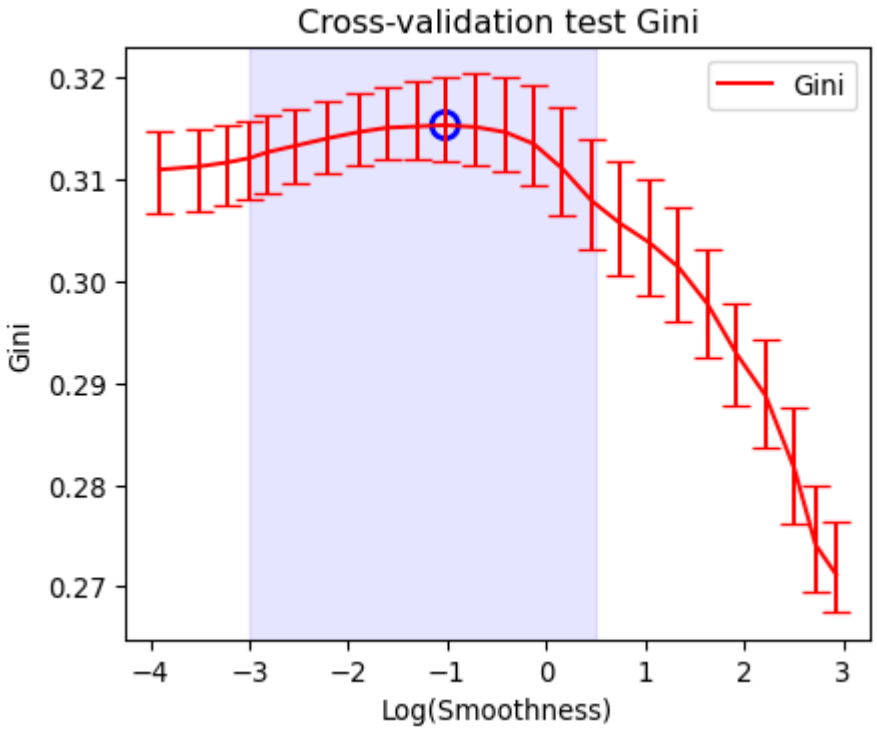
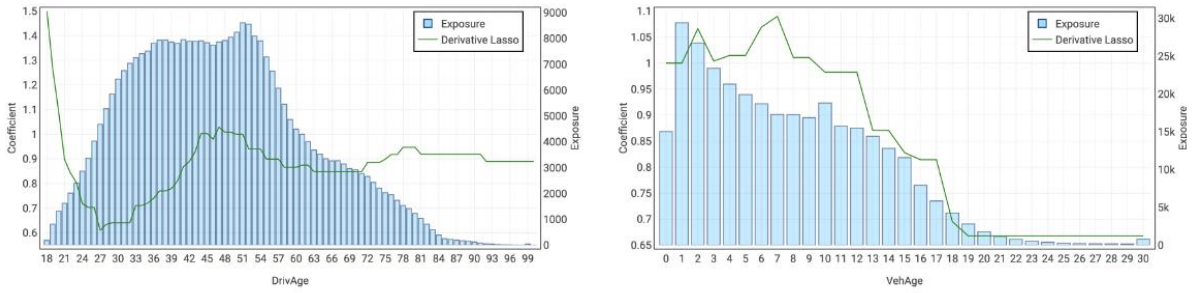



Figure 4.3: Cross-validation Gini results for a 4-fold cross-validation. The highlighted region represents models of approximately equal Gini performances. Notably, for Log(Smoothness) = -4, there is no significant difference between the penalized and unpenalized GLM models.

For penalized GLMs, the cross-validation procedure does not identify a single optimal smoothness parameter value but instead reveals a range of suitable smoothness values. The concavity (or convexity) of the performance graph can be theoretically explained through the classical bias-variance tradeoff of a model's generalization error [Hastie et al. (2009)]. Therefore, the modeler can safely assume that for all penalized GLMs, the cross-validation procedure will detect a single interval of interest for the smoothness parameter. Models within this range exhibit similar performance, making it challenging to purely rely on data-driven considerations to select one model over another. As a result, we recommend employing actuarial and subject matter expertise to make the final model selection in the subsequent expert-driven step.

## The expert-driven step

After identifying a smoothness interval that yields approximately equally performing models, the modeler should focus on inspecting the properties of the models within that range. For each smoothness value within the defined interval, we will examine a model fitted using the *entire modeling dataset* (refer to the expert-driven column of Figure 4.2). We begin the analysis by evaluating the model that achieves the highest Gini performance in absolute terms. Figure 4.4 showcase the coefficients for the driver age and vehicle age variables in this model.




*(a)* *(b)*

Figure 4.4: Coefficient plots for the driver age (panel a) and vehicle age (panel b) variables. The model is obtained by selecting the smoothness parameter value that maximizes the Gini score in cross-validation (value highlighted by a blue circle in Figure 4.3) and then fitting to the entire modeling dataset.

For this model, we observe that the derivative lasso penalty does influence the structure of the coefficients, resulting in some adaptive groupings. However, the quality of the model still remains questionable. While the overall shapes of the coefficients seem reasonable, we observe many local trend reversals or "wiggles." For example, small trend reversals can be observed for a driver age of 45, and for vehicle ages of 2, 7, and 8.

Even though the model shows the **best generalization performance** when evaluated on the test data in cross-validation, an expert-driven review may conclude that it **captures some noise and requires adjustments to make it actuarially sound**. We argue that the modeler *should not* try to remove these trend reversals by manual level grouping (for example, by combining vehicle ages of 1, 2 and 3 into a single level) and re-run the procedure, as one would typically do with an unpenalized GLM. Instead, we recommend selecting a model with a higher smoothness parameter value until the unexpected trend reversals disappear.

The rationale behind manual level grouping can be attributed to the lack of credibility in individual segments. In the context of penalized GLMs, the smoothness parameter allows for adjusting the credibility assigned to the data. When there is no clear explanation for the observed trend reversals, it suggests that the model is assigning too much credibility to the data. By choosing a model with a sufficiently high smoothness parameter value, it is expected that the trend reversals would disappear. In our example, the sufficiently high smoothness

level (and the corresponding model) is marked by a black circle in the grid search shown in Figure 4.5.

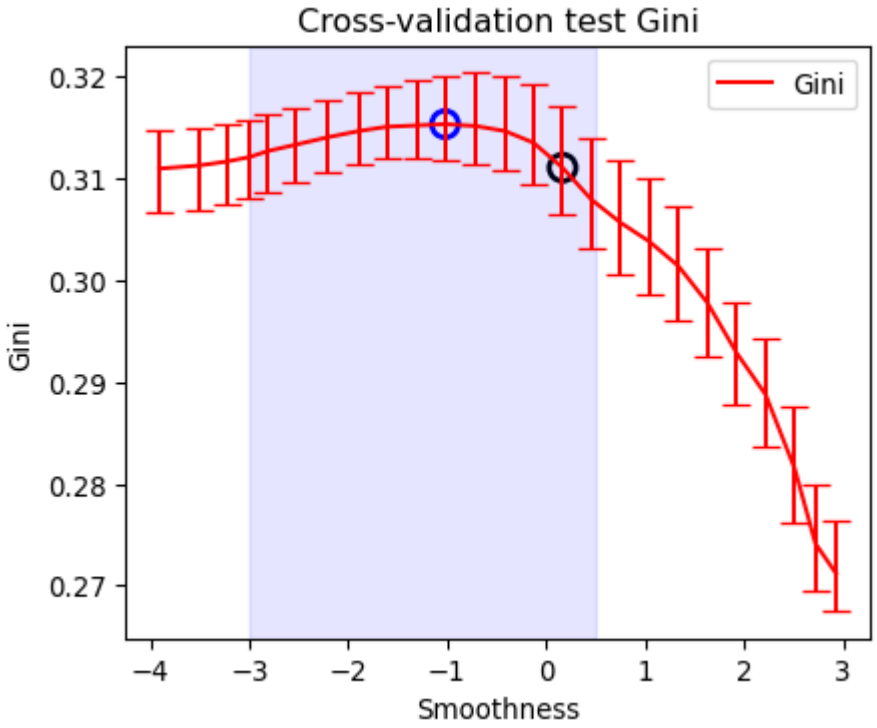
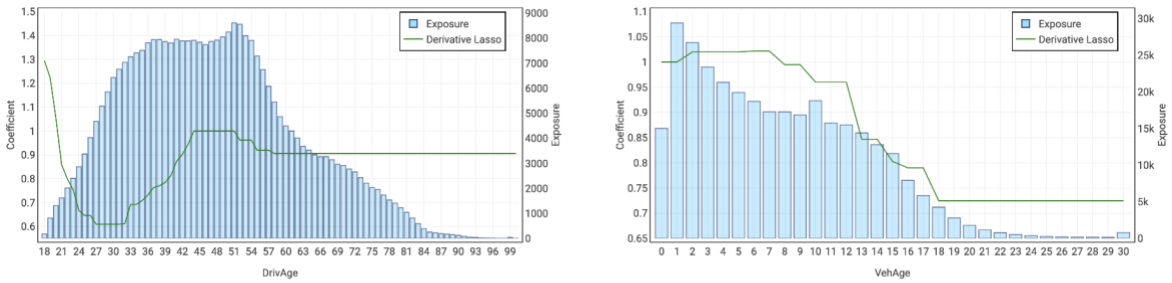



Figure 4.5: Same as Figure 4.3. The smallest smoothness parameter value that eliminates undesirable local trend reversals in the coefficient plots is marked with a black circle.

Figure 4.6 depict the coefficients for the driver age and vehicle age variables in this model. Visual analysis reveals that the model effectively captures the overall nonlinear trends in the data while avoiding undue sensitivity to local variations. The model not only strikes a perfect balance between robustness and expressiveness but is also *actuarially sound*.




*(a)* *(b)*

Figure 4.6: Equivalent figure to Figure 4.4 but with higher smoothness value (highlighted by a black circle in Figure 4.5). The increased smoothing removes undesired local trend reversals in the coefficients.

In the example using the freMtpl dataset, increasing the smoothness parameter is sufficient to finalize the model construction. In other scenarios, a potential complexity may arise where a trend is "smoothed" while it should not. A typical case of such behavior can be found in variables whose tails have low exposure. For example, Figure 4.7 depicts the behavior for an artificially constructed variable "Number of prior accidents" in auto insurance. The majority of policies exhibit either 0 or 1 accident, while a few policies have 2 or 3 accidents. The presence of more than one accident is a crucial risk indicator, so an ideal model should account for this trend. However, the derivative lasso model assigns the same risk level to individuals with 1 accident. At first glance, this might appear to be a limitation of the derivative lasso framework. Nevertheless, these scenarios are particularly interesting, as they highlight a key distinction in the drivers of modeling decisions between simple, unpenalized GLMs and derivative lasso models.

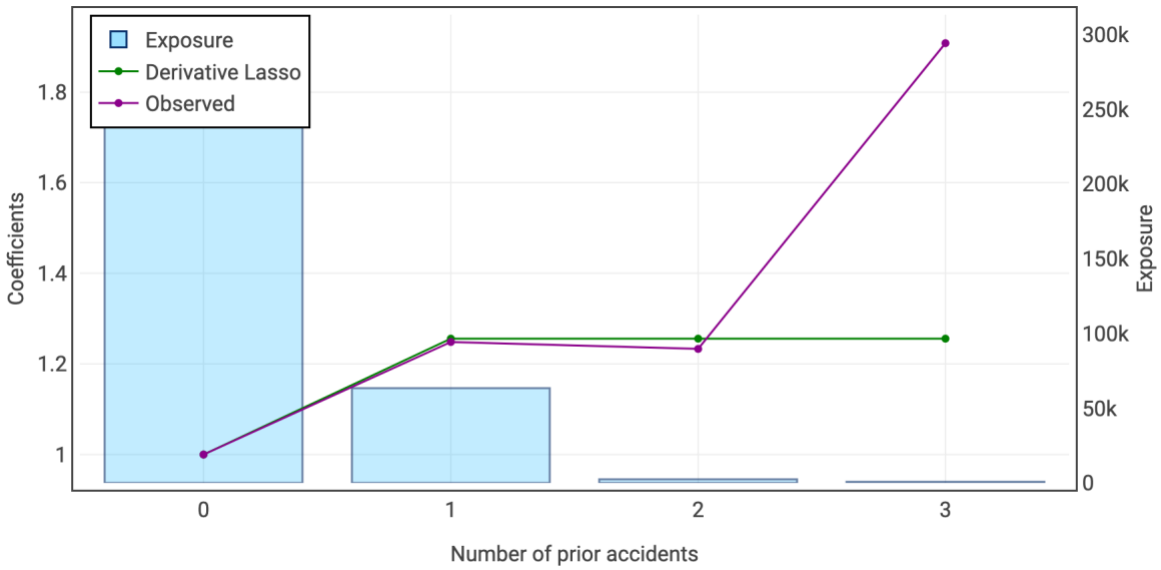



Figure 4.7: Comparison between derivative lasso coefficients and observed claims for the variable representing the number of prior accidents.

## Technical versus judgment-based decisions

The ability of penalized GLMs to control the amount of credibility given to the data implicitly highlights two main types of modeling decisions that the modeler should acknowledge:

- **Technical decisions:** modeling decisions that aim to improve the quality of the model's fit to the data.
- **Judgment-based decisions:** modeling decisions that aim to incorporate the modeler's opinions and experience, often based on judgment or business criteria.

In standard GLM modeling, there is no clear separation between these two families of decisions since feature engineering in GLMs achieves simultaneously technical and judgment-based goals.

A simple example of such overlap is the variable “Number of prior accidents” described above. To fit this variable, a modeler could choose either a log transformation or a linear transformation. Both of these transformations would produce similar results where there is sufficient data (for segments with 0 and 1 accidents) but would differ significantly where there is very little data (for segments with 2 or more accidents). The test statistics for each of these transformations are likely to be quite similar, as the contribution to the error for policies with 2 or more accidents is likely to be insignificant. **The model’s data is not guiding the indication for levels 2 or greater - it is the variable transformation alone**.

By selecting a log transformation, a modeler is essentially selecting a method of extrapolation for this factor. By selecting a linear transformation, a modeler is selecting a greater extrapolation than with log. This extrapolation is purely judgment based (since exposure is low), but it appears as if the model is technically justifying this extrapolation!

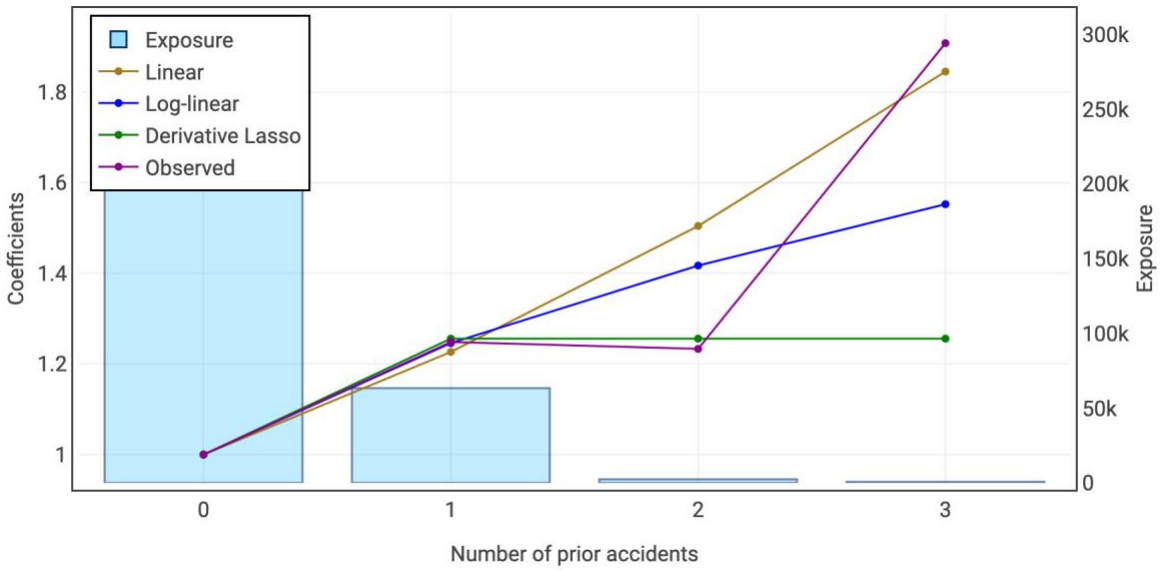



Figure 4.8: Comparison of coefficients result from fits with linear transformation Claim ~ Prior accident (yellow line), log transformation Claim ~ Log(Prior accident) (blue line) and the derivative lasso approach (green line), for the variable of number of prior accidents. Observed claims are represented by a purple line.

In unpenalized GLMs, feature engineering is a data-driven exercise only up to a point. Then, feature engineering begins to reflect the judgment of the modeler, who must decide which hypotheses, including variables and transformations, are tested or considered.

**The distinction between technical and judgment-based decisions is ambiguous in unpenalized GLMs, but it becomes well-defined in derivative lasso.** In derivative lasso, the chosen smoothness parameter governs the responsiveness of all factor curves. The smoothness automatically determines the placement of caps and floors based on the credibility of the data, free from human biases. Moreover, the smoothness parameter is unaffected by the modeler’s choices for variable transformations.

Using the derivative lasso methodology, the model will indicate a different factor for 0 and 1, but will not indicate extrapolation. The **extrapolation must be selected** and will **clearly be a judgment-based decision** instead of a technical decision. When we observe that levels with low exposure are grouped when they should not (false negative), this means that there is not enough credibility to the data to justify that expected trend. The derivative lasso offers a non-biased view to the data and decisions with lack of data should always be complemented using actuarial judgment. Fortunately, the model structure of both unpenalized and penalized GLM allows us to easily complement the model with actuarial judgment as we now show.

## 4.3 Incorporating judgment into penalized GLMs

Applying judgment in the context of GLMs is the ability to manually edit the structure of the model. In the previous example of the "Number of prior accidents" Figure 4.8, this means manually interpolating the impact of the coefficients on levels 2 and 3 as these levels cannot be reliably estimated due to the lack of the data.

Both unpenalized and penalized GLMs make the incorporation of judgment possible thanks to their clear output structure. Any GLM estimate can be expressed as a multiplication of individual rating factors, one per each variable in the model plus a base. Figure 4.9 showcases an extract of a penalized GLM model consisting of 3 variables (Area, Vehicle Gas, Vehicle Power) together with the resulting prediction of a policy living in area D, with Regular gas and vehicle power equal to 9.

The ability to develop rate-table-based models, other than simplifying real time IT implementation of insurance rating, allows for easy application of judgment by means of modifying the individual GLM factors.

| Variable | Level 1 | Factor Indicated |
|---|---|---|
| Base | Base | 500 $ |
| Area | A | 0.91 |
| Area | B | 0.91 |
| Area | C | 1.00 |
| Area | D | 1.05 |
| Area | E | 1.10 |
| Area | F | 1.17 |
| VehGas | Diesel | 1.17 |
| VehGas | Regular | 1.00 |
| VehPower | 4 | 0.93 |
| VehPower | 5 | 0.98 |
| VehPower | 6 | 1.00 |
| VehPower | 7 | 1.01 |
| VehPower | 8 | 1.01 |
| VehPower | 9 | 1.13 |
| VehPower | 10 | 1.13 |
| ... | ... | ... |

**Policy**

Area = D
Vehicle Gas = Regular
Vehicle Power = 9

**Premium ($) = 500 * 1.05 * 1.00 * 1.13**

Figure 4.9: Table-like representation of the model, together with the resulting prediction of a policy living in area D, with Regular gas and vehicle power equal to 9.

To be able to apply such selections it is necessary to be able to naturally separate the contribution of each individual input variable to the overall prediction. Such a separation must be clear so that the modeler can fully understand the impacts of the application of actuarial judgment.

We argue that being able to represent a model as a table-based output is a necessity for this application of judgment. Some modeling techniques can incorporate mathematical constraints which can be driven by judgment-based considerations (for example, forcing to follow a monotonically increasing trend[10]). However there will always be the need to potentially deviate from the final model to satisfy judgment-based considerations. For these applications, the modeler must have a transparent mean to apply these edits which are satisfied by the rate-table structure. For modeling techniques which do not exhibit a table-like structure (such as GBMs), applying these deviations is impossible.

Care must be taken when applying actuarial edits to the GLM tables. A basic requirement after a coefficient edit would be to recompute a new value for the base (or equivalently, intercept) of the model such that the average of the predicted values before the edit matches the one after the edit. Depending on the use-case, the modeler may decide to include the resulting edit as an offset to the model (fixing the coefficient) and refitting the resulting model. There are several approaches that are equally applicable for both standard and penalized GLMs to robustly incorporate the judgment of the modeler.

## 4.4 Conclusion

This chapter showcased and motivated a process to build appropriate GLM models in a two-step procedure:

1. **Data-driven step**: from a coarse grid of smoothness parameters, use cross-validation to assess and define a restricted interval of smoothness parameters which yields models that generalize correctly to unseen data.
2. **Expert-driven step**: within the restricted interval, fit models on the whole modeling dataset. Perform a visual analysis of the model, in accordance with GLMs best practices. The modeler should define the level of credibility to give to the data through analysis of the coefficients. Whenever dubious trend reversals are observed, the smoothness parameter shall be increased until such reversals are removed. There are cases where the lack of data for given segments should be compensated with application of actuarial judgment, and this judgment is easily applicable due to the inherent rate-table structure of the model.

[10] Penalized GLM can also follow non decreasing trends by incorporating constraints in the model (also known as isotonic regression). The definition and treatment of these constraints is outside of the scope of this publication.

# 5. Comparison with other modeling techniques

In the toolkit of predictive modelers, various modeling techniques are available beyond standard or penalized GLMs. One such method is Generalized Additive Models (GAM), a popular approach capable of automatically modeling nonlinearities in insurance data. This chapter will highlight the differences and similarities between GAMs and penalized GLMs, specifically the GLMs with lasso and derivative lasso penalties.

## 5.1 Black-box explainability versus table-based output

There are several reasons for which GLMs are the industry standard for insurance modeling, and one of them is its rate-table-based output. The rate-table structure (see Section 4.3), shared by both GLMs and penalized GLMs, provides three key benefits[11]:

1. **Explainability:** Rate tables by design allow to separate the contribution of each individual input variable to the overall prediction, allowing to describe the impact of each separate variable to the estimates of the model.
2. **Judgment** application: The explicit format allows to let the model deviate from the estimates of the model.
3. **Implementation:** Rate tables' explicit formulas simplify the integration with IT systems such as rating engines and reduce costs compared to other modeling techniques (e.g., tree-based techniques).

The first point, **explainability**, is the main objective of XAI, the acronym for Explainable Artificial Intelligence, a rapidly growing field within the domain of machine learning. One of the research areas of XAI is interpretable machine learning, which is a useful umbrella term that captures the "extraction of relevant knowledge from a machine-learning model concerning relationships either contained in data or learned by the model". There are several methodologies such as partial dependence plots and Shapley values that, given as input a black-box model such as tree ensembles (GBMs) or a deep neural network, can provide an approximate description of how each variable impacts the model.

These metrics provide a powerful tool to better understand the output of arbitrary machine learning models. However, while they may help or support explainability of the model, they do not allow the application of judgment, nor are compatible with some insurance IT systems such as rating engines.

[11] The list can be expanded by considering the ability to justify what the models are doing, for example, to regulators or to verify fairness of the model. The ability to monitor the quality of the model: by understanding how the model are making decisions, we are then able to debug when thing go wrong more easily.

## 5.2 Generalized Additive Models (GAMs)

### The GAM formula

Generalized Additive Model or GAMs are a modeling technique popularized by the seminal paper by Hastie and Tibshirani (1986), where, for the first time, the prediction of a GLM was expressed in a functional formula:

$$\text{GAM}(x_i) = g^{-1}\left(\beta_0 + \sum_{j \in \text{Variables}} f_j\left(x_{ij}\right)\right).$$

Here $g^{-1}$ represents the inverse of the link function (e.g., exponential function), $x_i$ represents the policy variable for policy $i$, and $j$ is the index for the $j$-th variable in the policy (e.g., driver age, vehicle brand).

The main contribution of the GAM formula is the explicit decomposition of the effect of each variable from a linear form (as in a GLM) to an arbitrary nonlinear representation $f$.

The GAM representation explicitly shows the contribution of each individual variable to the overall prediction of a GAM. This representation provides a natural rate-table representation. Both GLMs and penalized GLMs can be expressed using the more generic GAM formula.

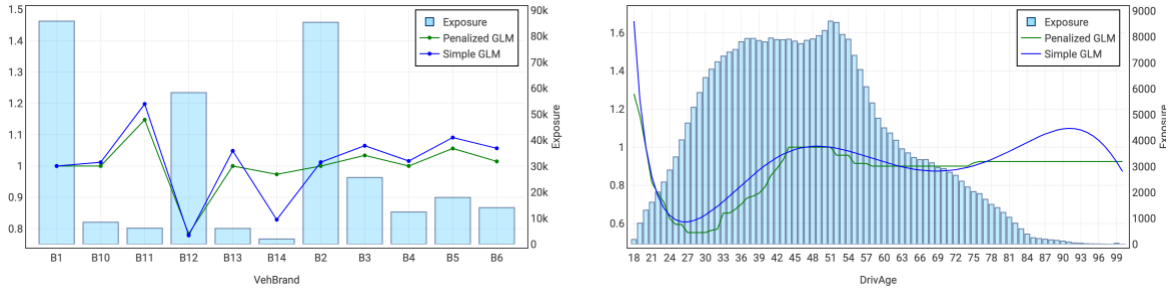



Figure 5.1: Both unpenalized and penalized GLM do have a GAM representation. For both models, the GAM function plot is represented by the blue line for unpenalized GLM and the blue line for penalized GLM.

We illustrate the connections among these methods by showing the GAM formula for both the multivariate unpenalized and penalized GLMs (presented in Section 1.3 and Section 3.4, respectively).

In the case of the vehicle brand variable, in both unpenalized and penalized GLMs, an individual coefficient was assigned to each level due to the categorical nature of the variable. For example, $f_{\text{VehBrand}}(B\ 14) = \beta_{B\ 14}$ . We can express mathematically this generic relationship via the formula

$$f_{\text{VehBrand}}(x) = \sum_{j \in B1,\ldots,B14} \mathbb{1}_{x=j}\ \beta_j.$$

Where the $\mathbb{1}_{x=j}$ is the indicator function that returns 1 if $x$ is equal to the relative vehicle brand, and 0 elsewhere.

For the driver age variable, the parametrization differs from unpenalized GLMs to penalized GLMs. In the case of derivative lasso, where each level has its own coefficient, the GAM formula still is

$$f_{\text{DriverAge}}(x) = \sum_{j \in 18,\dots,99} \mathbb{1}_{x=j} \ \beta_j.$$

For the simple, unpenalized GLMs, we provided an example where the driver age was modeled via the relationship ClaimNb ~ DrivAge + I(DrivAge**2) + I(DrivAge**3) + I(DrivAge**4) (see Figure 5.1). The corresponding GAM representation is

$$f_{\text{DriverAge}}(x) = \sum_{j \in 1,\dots,4} \beta_j \, x^j,$$

with $\beta_j$ being the linear coefficient for the $j$-th-degree term of the polynomial.

Since **derivative lasso can be represented via the GAM formula**, and it explicitly decomposes the contribution of each variable to the model, one can define derivative lasso solutions to be called GAMs.

The GAM formula and associated representation are very generic and powerful. This flexibility makes it hard to define when, for example, a GLM becomes a GAM or vice versa. The reader should as well be aware that similar modeling techniques may have different names as described in Molnar (2020).

### The "spline" GAMs

While the GAM formula and associated representation are very generic, in statistical literature the term GAMs is usually used to indicate models that use the very specific family of **spline** functions.

In general, a polynomial GLM of degree $k$ can be represented via the formula $f(x) = \sum_{j=1}^{k} \beta_j \, x^j$. Figure 1.9 illustrated how polynomials may exhibit significant instabilities, in particular at the tail of the distribution. The idea of splines is to replace each unstable, $j$-th-degree term $x^j$ of the polynomial with a more generic **spline function** $\phi_j(x)$. These functions $\phi_j$ have mathematical properties more desirable than polynomials, and limit (but do not completely eliminate) the numerical instability of standard polynomials.

A spline function can capture nonlinearities in the same fashion as a polynomial transformation would, but in a **more stable** way due to their desirable mathematical properties. The modeler should be aware that there are many different families of splines (e.g., the open-source package *mgcv* provides 11 of such families). Figure 5.2 provides a representation of the cubic spline family, one of the most explicit splines. The user should be

aware that the default spline basis in *mgcv* is thin plate splines, and their visual representations are less intuitive than the cubic splines. We will not go into details of the various classes of splines, as their functional representations require a cumbersome mathematical notation. Luckily, the availability of softwares such as the *mgcv* R package allows to easily switch among various spline bases.

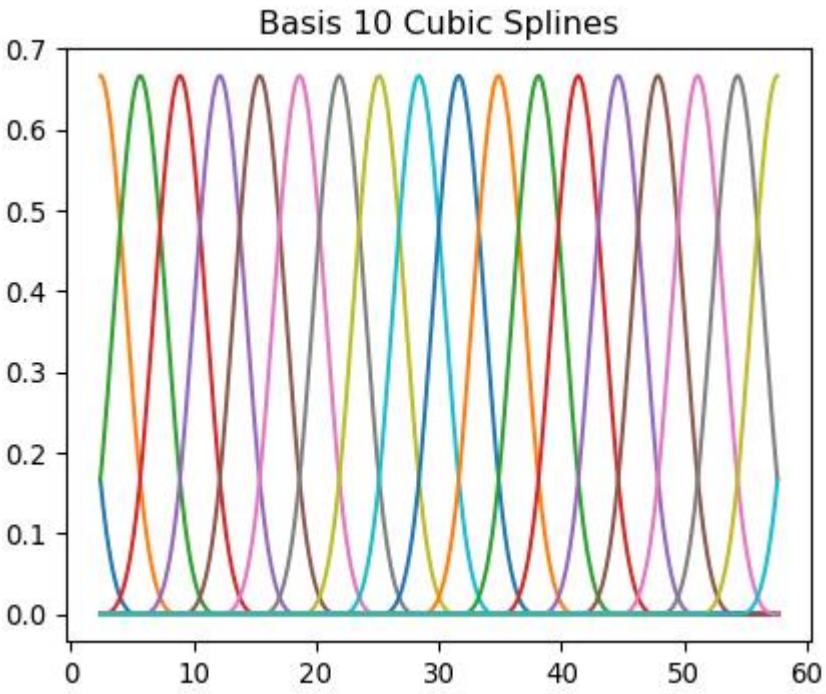



Figure 5.2: Plot of 10 cubic spline basis functions at 10 equidistant knots.

Spline-based GAMs are able to model highly nonlinear effects. We consider data from Venables and Ripley (2013), representing a series of measurements of head acceleration in a simulated motorcycle accident. To fit a spline GAM to the data, the user first needs to decide the number of splines to include in the model. The number of splines has to be defined for each variable in the model, similarly to the definition of the degree of a polynomial. Usually these splines are situated on equi-spaced intervals within the range of numeric values assumed by the variable of interest, even if, a priori, the modeler could choose different knot positions.

Figure 5.3 shows the fit of a GAM with 20 cubic splines. The overall GAM fit consists of the combination of the 20 spline basis functions and 20 coefficients $\beta$. The 20 coefficients can be computed using the standard maximum likelihood procedure (as for a GLM). The right-hand side of the graph depicts how all the 20 different spline factors contribute to the final effect of the GAM (blue line). By qualitatively looking at the data (represented by the gray scatterplot), we can verify that the nonlinearities are well captured. The left-hand side of the figure compares the GAM model with the observations of the data.

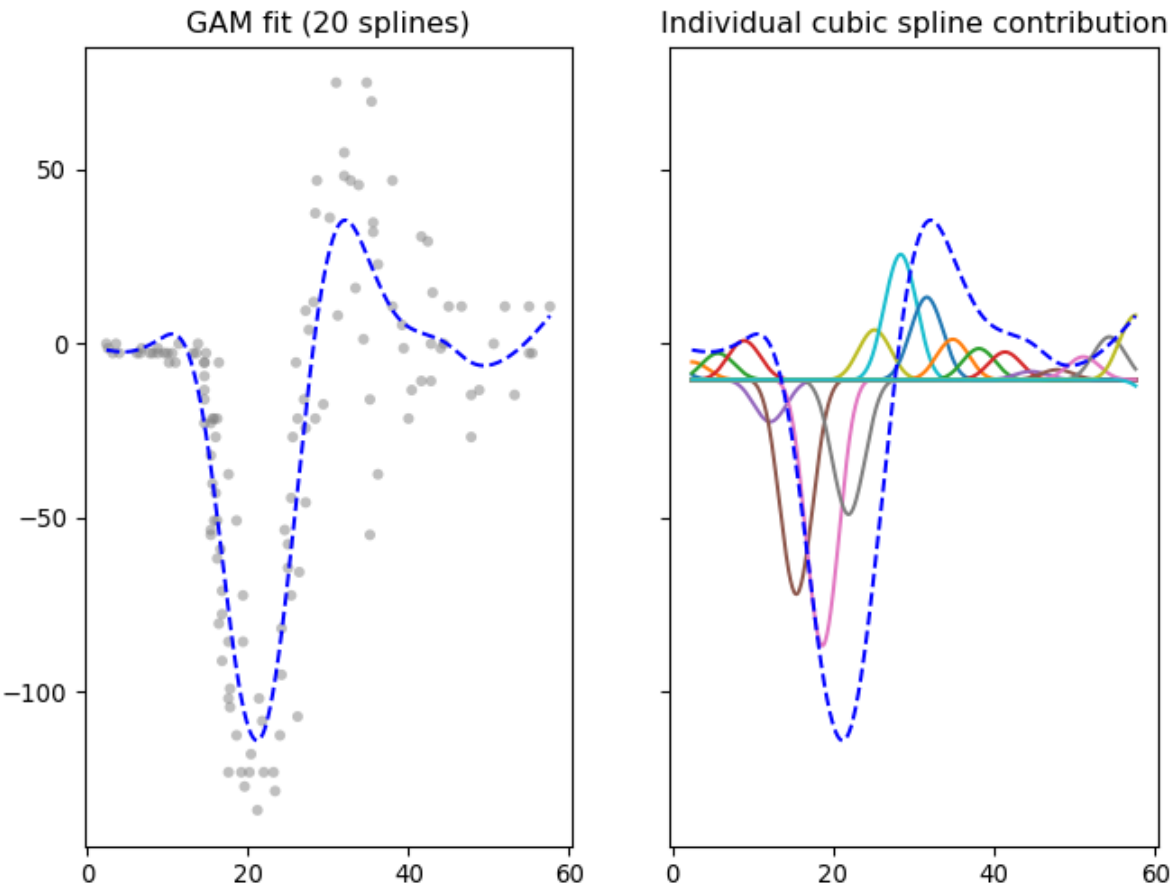



Figure 5.3: Results of fitting a GAM using 20 cubic spline functions to the motorcycle data. Left panel shows the total predictions of the GAM fit along with the observations. Right panel shows the decomposition of the total predictions into contributions from individual spline terms.

Since splines are, by definition, a type of feature transformation, they require the user to specify some decisions for each variable of interest - in the same way that a modeler should choose the degree of a polynomial in a classic GLM (see Section 1.2.4 for an application of polynomial transformation for the variable Vehicle Age).

Examples of these parameters include:

- **Type of splines** : the R package *mgcv* offers 11 different flavors of spline bases, with the default being the low-rank thin plate spline basis. In practice, the choice of the specific splines has little material impact on the results of the obtained GAMs.
- **Number of splines** : the number of underlying spline functions to be considered in the fit. A small number of splines may lead to underfitting, and a large number may lead to "wiggly" functions overfitting the data. In the case of cubic splines, this corresponds also to the number of "knots" for the individual spline terms.

Depending on the complexity of the nonlinearities that the user wants to describe, the user should define the number of basis splines **for each variable in the model**. The way that this parameter is chosen is similar to how the GLM decisions are made in the modelling process.

### GAM model selection

GAMs provide an adaptation of simple GLMs by using more sophisticated variable transformations than standard polynomials, such as splines. Even if they exhibit better mathematical properties, the model complexity brought by the splines needs to be closely monitored by the modeler.

Figure 5.4 showcases how different numbers of splines impact the contribution of the GAMs.

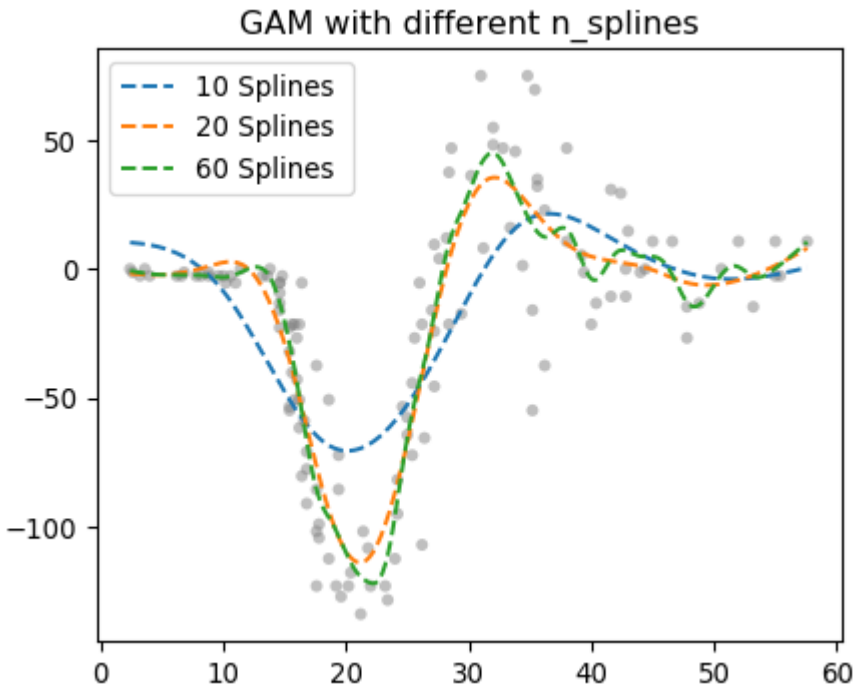



Figure 5.4: Results of fitting GAMs using different numbers of splines.

GAMs offer two ways to control model complexity and mitigate overfitting:

- **Tuning the number of splines:** As shown in the figure above, the higher the number of splines, the higher the probability of overfitting. When overfitting is detected (as seen in the case above, with the "wiggling" nature of the 60-spline model), the modeler can decrease the number of splines.
- **Applying penalization:** The standard GAM package *mgcv* provides the option to apply a ridge penalty to the individual $\beta_j$. The penalty influences the "amplitude" of the individual spline functions. A higher value of the penalty value will shrink individual splines and thus create a less "wiggling" aggregate factor curve.

The choice of the penalty value $\lambda$ in GAMs can be determined via cross-validation (as described in Section 4.2.1) or via some more GAM explicit criteria (that do not translate, for example, with lasso or derivative lasso penalizations), such as AIC optimization or the Bayesian approach of the restricted maximum likelihood.

There are several ways to monitor the quality of the model and to ensure the model does not overfit:

- A high **concurvity** occurs when some smooth term in a model could be approximated by one or more of the other smooth terms in the model. This can occur for example when two correlated variables are overparameterized (that is, have an unnecessarily high number of splines). In the context of GAM splines, models can suffer from high concurvity on top of high collinearity. That is problematic for the GAM as the complex shape of the splines may be over sensitive to spurious correlation, leading to estimates that are non-stable. Packages such as *mgcv* provide different types of concurvity by computing how much the impact of a given variable to the model can be represented by the other variables. This metric has a numeric value between 0 and 1, and variables having concurvity above 0.8 should be questioned if not removed.
- **P-values** and **confidence intervals** can also be visualized per variable in a GAM. Assuming no ridge penalization applied to the coefficients of the GAM, the GAM is simply a GLM with spline feature engineering. For this reason, it is still possible to compute p-values for each individual coefficients of the GAM, and then aggregate these statistics to a per variable visualization. Under the hood, the **mgcv** software will compute the second order derivative of the likelihood and use it as an input to a formula that will return either P-values or confidence intervals. When penalization is applied, care must be taken to interpret the results. The ridge penalty $|\beta|^2$ is differentiable, so it is still possible to compute the second order derivative of the likelihood of a penalized GAM. This second order derivative can be still used as an input to the formula to compute P-values or confidence intervals. The result will not be a "real" p-value from a statistical theory prespective, but the approximation can still be useful to the modeler in the review of the model. Since the lasso is not differentiable, the same approximation can not be applied.

### Comparison with derivative lasso

Both derivative lasso and spline GAMs can be expressed via the GAM formula. Both methodologies can be expressed as a combination of functional basis and use penalized regression, but have key differences that we now describe.

#### *Basis representation*

Spline GAMs introduce a new perspective on how to approach modeling: the basis representation of the effect and the usage of the penalization to control complexity. It is possible to use this analogy in the context of the derivative lasso.

Derivative lasso outputs functions which are stepwise constant. A functional basis that can represent a piecewise constant interpolation is depicted in Figure 5.5. The fit using stepwise functions may be more robust due to the particularly noisy nature of insurance data or the preference of actuaries of binning. In other words, step functions serve as a more robust basis set in the context of insurance risk modeling.

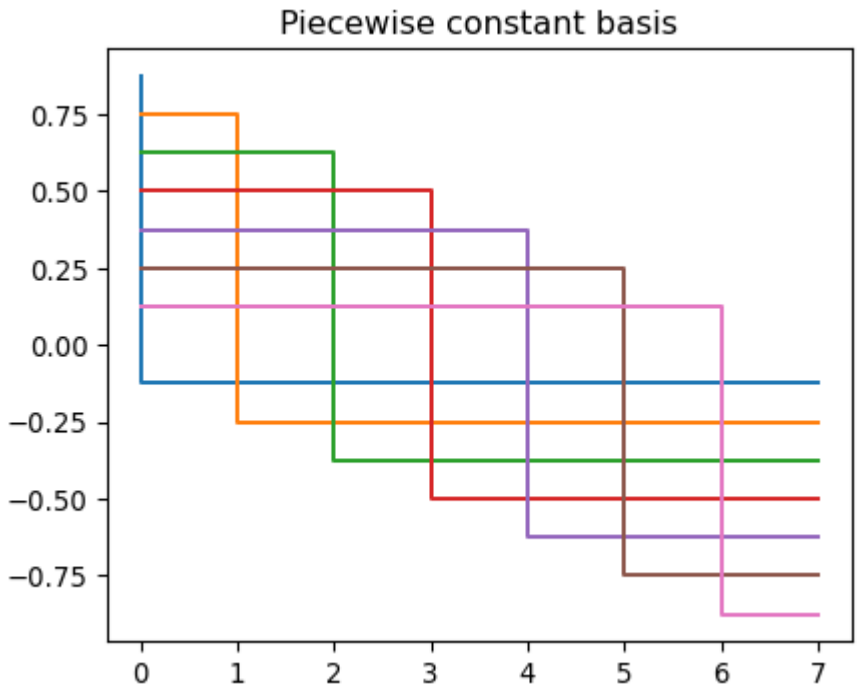



Figure 5.5: Visual representation of sampled piecewise constant basis functions.

*Role of the penalty*

Spline GAMs are often combined with ridge penalty for the sole purpose of controlling overfitting. The penalty will shrink the coefficients of a GAM (which control the amplitudes of the individual spline functions) resulting in a final overall contribution that tends to be more stable around zero.

Similarities and differences between spline GAMs and derivative lasso can be directly visualized in Figure 5.6.

In this specific database, the two fits are materially the same. The GAM has some swoops or curves that do not seem justified both in the right tail and in the left tail.

In this example, an appropriate GAM model has no material differences from an appropriate derivative lasso model. There are theoretical results that favor the derivative lasso over a spline GAM methodology. It can be proven, both empirically and theoretically, that derivative lasso adapts to the local level of smoothness much better than smoothing splines. Remarkably, the derivative lasso provides the **optimal** constant wise interpolation as it converges to the oracle solution for a wide class of functions [Tibshirani (2014)].

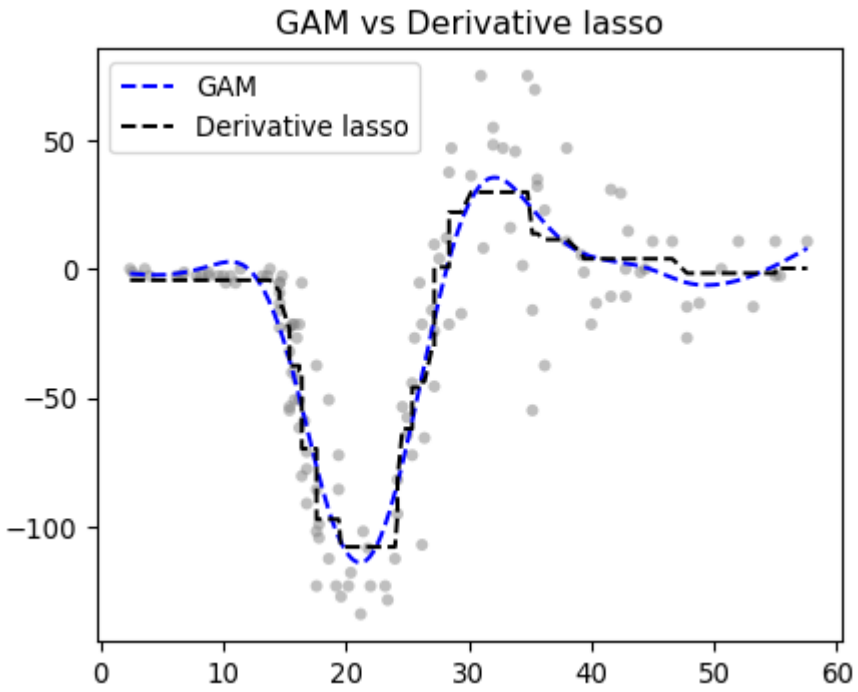



Figure 5.6: Comparison between GAM and derivative lasso coefficients in the motorcycle data.

*Main parameters*

Fitting a spline GAM requires one to select several parameters prior to the computation. For example, the nature of the GAM requires choosing the number of splines and the spline type **for each variable in the model**. In addition, a ridge penalty parameter may be required to be tuned. Conversely there is only **one single parameter** for both lasso and derivative lasso. This single parameter is used to penalize all variables in the model and create factor curves simultaneously by evaluating the credibility of the data in the methodology outlined below. Because of this, model building and evaluation in derivative lasso are greatly simplified.

*Model auditability*

In both methodologies the quality of a model can be analyzed by looking at the variable graphs, to verify that the estimated trends are appropriate. However, we now argue that reviewing ad-hoc GAM metrics such as concurvity and p-values is not an appropriate to evaluate the quality of a derivative lasso model. Computing **concurvity** on derivative lasso will provide very pessimistic estimates of the quality of the model, which makes the concurvity metric not actionable. Concurvity formulas in packages such as *mgcv*, are designed for complex spline representations such as the cubic spline basis in Figure 5.2.

The cubic spline representation brings much more complexity to the model than the robust constant wise basis used in derivative lasso. Furthermore, in derivative lasso there are as many parameters (and hence, basis functions) as levels in the variable. In this representation, the penalty plays a central role to provide the structure of the resulting model. Conversely, in GAMs it is strongly not recommended to have as many splines as variable levels in a function

to avoid overfitting. When computing concurvity metric on derivative lasso, the formulas will interpret the model as having as many splines as the number of levels, which will significantly overestimate the over-fitting potential of an otherwise stable model.

For example, the most parsimonious estimate of concurvity of *mgcv*, computed on any derivative lasso solution, should return a very similar concurvity of a "saturated" spline, which has as many parameters as factors (as shown in Figure 5.7).

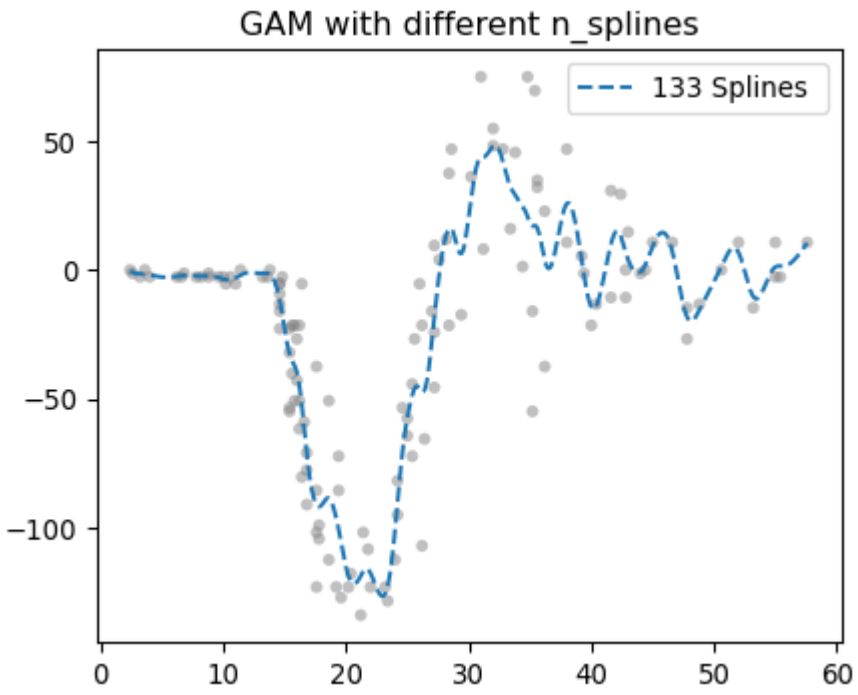



Figure 5.7: 'Saturated' GAM with as many splines as data points. A parsimonious estimate of concurvity of any derivative lasso solution will be similar to the overfitted saturated GAM model.

Anecdotally, one can say that computing concurvity in derivative lasso is not appropriate because it has no curves.

We also note that packages such as *mgcv* provide, among model diagnostic, an approximated value for the AIC. As the lasso problem can be seen as an approximation to the minimization of the AIC formula (see discussion at the end of Section 4.1), it is not appropriate to evaluate the AIC on derivative lasso as a metric that theoretically approximates the out-of-sample deviance of the model. In both GAMs and derivative lasso, the AIC can be effectively replaced by the most generic cross-validation procedure described in Section 4.2.1.

## Conclusion

This paper demonstrates that GLMs, when applied with lasso and derivative lasso, can effectively *model nonlinear relationships* and *manage low-credibility segments*. We have shown how lasso is suitable for categorical variables, and how derivative lasso natively captures the nonlinear effects observed in ordinal variables. The modeler retains full control over the impact of the penalties by adjusting the value of the $\lambda$ parameter, which acts as a global credibility standard assigned to the data.

The approach marks a substantial departure from the traditional GLM methodology, which generally follows a two-step process. The first step defines an hypothesis of the GLM structure, such as manual variable selections, transformations, and feature engineering. The second step consists of fitting and assessing the quality of the model. This assessment can only evaluate the combined effect of the feature and its engineering and cannot differentiate the two. If a variable is predictive, but the feature engineering is poor, the assessment will show that the coefficient is insignificant. When the assessment falls short (which is often the case), the modeler must return to the first step and repeat. For this reason, the standard GLM approach necessitates a highly iterative process that involves human intervention at every step.

The derivative lasso framework provides a solution that significantly reduces the number of manual iterations by combining the two steps mentioned above into one single process. In particular, the "feature engineering" is determined entirely by the data *during* derivative lasso model fitting and does not depend on a modeler's correct interpretation and feature engineering of the data. The single step of tuning the smoothness parameter creates more or less complex models taking into account the effect of all other variables. Such an automatic fitting process comes with a native table-based format that can be readily visualized and deployed to production in a similar way than a GLM. Actuarial judgment is applied via the choice of the smoothness parameter and factor selection in the expert-driven phase of model review.

Derivative lasso provides a transparent, sound, and fast solution for the application of predictive modeling in actuarial practice:

- First, it resolves the false dichotomy discussed in the introduction, where modelers need to choose between transparency (e.g., GLMs) and automation (e.g., GBMs).
- Secondly, by reducing the time required to develop models, lasso and derivative lasso allows the application of predictive modeling to other use-cases that may not have been considered before. For example, it is possible to easily define model updates, as one can couple both a complement of credibility (the current model or other best estimate) and the ability to detect nonlinearities.

- Third, the “cost of entry” is significantly reduced due to the simplicity of the modeling process and the separation of technical and judgmental decisions. A simple, credibility-based approach allows more junior modelers to build supportable models. The separation of technical and judgmental decisions combined with the transparent output allows highly skilled non-modeling actuaries to confidently make full use of their skills by adjusting the models to fully incorporate business knowledge.

We hope that this paper has familiarized the reader with the usage of derivative lasso techniques by providing both a mathematical foundation and an actuarial perspective. These two items combine to create a robust **modeling framework** that can be used to determine the best model for a wide range of actuarial applications.

# References


Anderson, Duncan, Sholom Feldblum, Claudine Modlin, Doris Schirmacher, Ernesto Schirmacher, and Neeza Thandi. 2007. “A Practitioner’s Guide to Generalized Linear Models—a Foundation for Theory, Interpretation and Application.” Towers Watson.

Casotto, Mattia, Marco Banterle, and Guillaume Beraud-Sudreau. 2020. “Credibility and Penalized Regression.”

Charpentier, Arthur, ed. 2014. *Computational Actuarial Science with R*. New York: Chapman; Hall/CRC. https://doi.org/10.1201/b17230.

Fujita, Suguru, Toyoto Tanaka, Kenji Kondo, and Hirokazu Iwasawa. 2020. “AGLM: A Hybrid Modeling Method of GLM and Data Science Techniques.”

Goldburd, Mark, Anand Khare, Dan Tevet, and Dmitriy Guller. 2016. “Generalized Linear Models for Insurance Rating.” *Casualty Actuarial Society, CAS Monographs Series* 5.

Hastie, Trevor, and Robert Tibshirani. 1986. “Generalized Additive Models.” *Statistical Science* 1 (3): 297–310. https://doi.org/10.1214/ss/1177013604.

Hastie, Trevor, Robert Tibshirani, Jerome H Friedman, and Jerome H Friedman. 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Vol. 2. Springer.

Hastie, Trevor, Robert Tibshirani, and Ryan Tibshirani. 2020. “Best Subset, Forward Stepwise or Lasso? Analysis and Recommendations Based on Extensive Comparisons.”

Jewell, William S. 1974. “Credible Means Are Exact Bayesian for Exponential Families.” *ASTIN Bulletin: The Journal of the IAA* 8 (1): 77–90.

Klinker, Fred. 2010. “Generalized Linear Mixed Models for Ratemaking: A Means of Introducing Credibility into a Generalized Linear Model Setting.” In *Casualty Actuarial Society e-Forum, Winter 2011 Volume 2*.

Leahey, Erin. 2005. “Alphas and Asterisks: The Development of Statistical Significance Testing Standards in Sociology.” *Social Forces* 84 (1): 1–24.

Miller, Hugh. 2015. “A Discussion on Credibility and Penalised Regression, with Implications for Actuarial Work.” *Actuaries Institute*.

Molnar, Christoph. 2020. *Interpretable Machine Learning*. Lulu. com.

Noll, Alexander, Robert Salzmann, and Mario V Wuthrich. 2020. “Case Study: French Motor Third-Party Liability Claims.” *Available at SSRN 3164764*.

Schelldorfer, Jürg, and Mario V Wuthrich. 2019. “Nesting Classical Actuarial Models into Neural Networks.” *Available at SSRN 3320525*.

Semenovich, Dimitri. 2013. “Applications of Convex Optimization in Premium Rating.” In *Casualty Actuarial Society e-Forum, Spring 2013*.

Tibshirani, Ryan J. 2014. “Adaptive Piecewise Polynomial Estimation via Trend Filtering.”

Venables, William N, and Brian D Ripley. 2013. *Modern Applied Statistics with s-PLUS*. Springer Science & Business Media.

Werner, Geoff, and Claudine Modlin. 2010. “Basic Ratemaking.” In *Casualty Actuarial Society*, 4:1–320.

Wüthrich, Mario V, and Michael Merz. 2023. *Statistical Foundations of Actuarial Learning and Its Applications*. Springer Nature.